

ACUTE RESPIRATORY DISTRESS SYNDROME (ARDS)

DESCRIPTION

Timing:	Within 1 week of known clinical insult or new or worsening respiratory symptoms
Chest imaging:	Bilateral opacities – not fully explained by effusions, lobar/lung collapse, or nodules
Origin of edema:	Respiratory failure not fully explained by cardiac failure or fluid overload. Need objective assessment (e.g., echocardiography) to exclude hydrostatic edema if no risk factor present
Oxygenation:	
Mild	$200 \text{ mm Hg} < \text{PaO}_2/\text{FIO}_2 < 300 \text{ mm Hg}$ With PEEP or CPAP $\geq 5 \text{ cm H}_2\text{O}$
Moderate	$100 \text{ mm Hg} < \text{PaO}_2/\text{FIO}_2 < 200 \text{ mm Hg}$ With PEEP $> 5 \text{ cm H}_2\text{O}$
Severe	$\text{PaO}_2/\text{FIO}_2 < 100 \text{ mm Hg}$ with PEEP or CPAP $\geq 5 \text{ cm H}_2\text{O}$

ELEMENT VALUES

1. Yes
2. No

ADDITIONAL INFORMATION

- Onset of symptoms began after arrival to your ED/hospital.
- A diagnosis of ARDS must be documented in the patient's medical record.
- Consistent with the 2012 New Berlin Definition.

DATA SOURCE HIERARCHY GUIDE

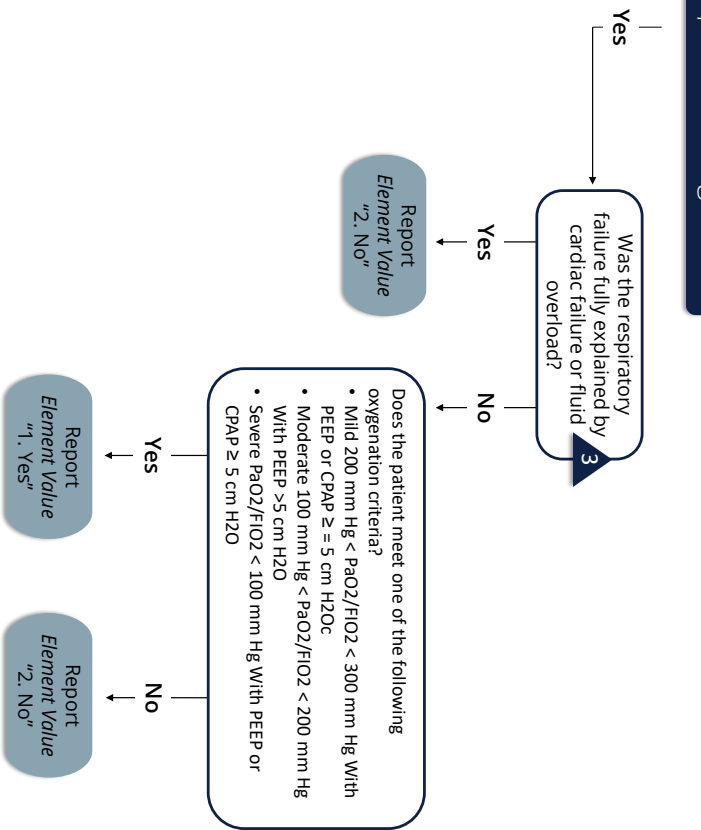
1. History and Physical
2. Physician Notes/Flow Sheet
3. Progress Notes
4. Case Management/Social Services Notes
5. Nursing Notes/Flow Sheet
6. Triage/Trauma Flow Sheet
7. Discharge Summary

ASSOCIATED EDIT CHECKS

Rule ID	Level	Message
18601	1	Value is not a valid menu option
18603	2	Element cannot be blank
18604	2	Element cannot be "Not Applicable"
18640	1	Single Entry Max exceeded

Acute Respiratory Distress Syndrome (ARDS) (pg. 2 of 2)

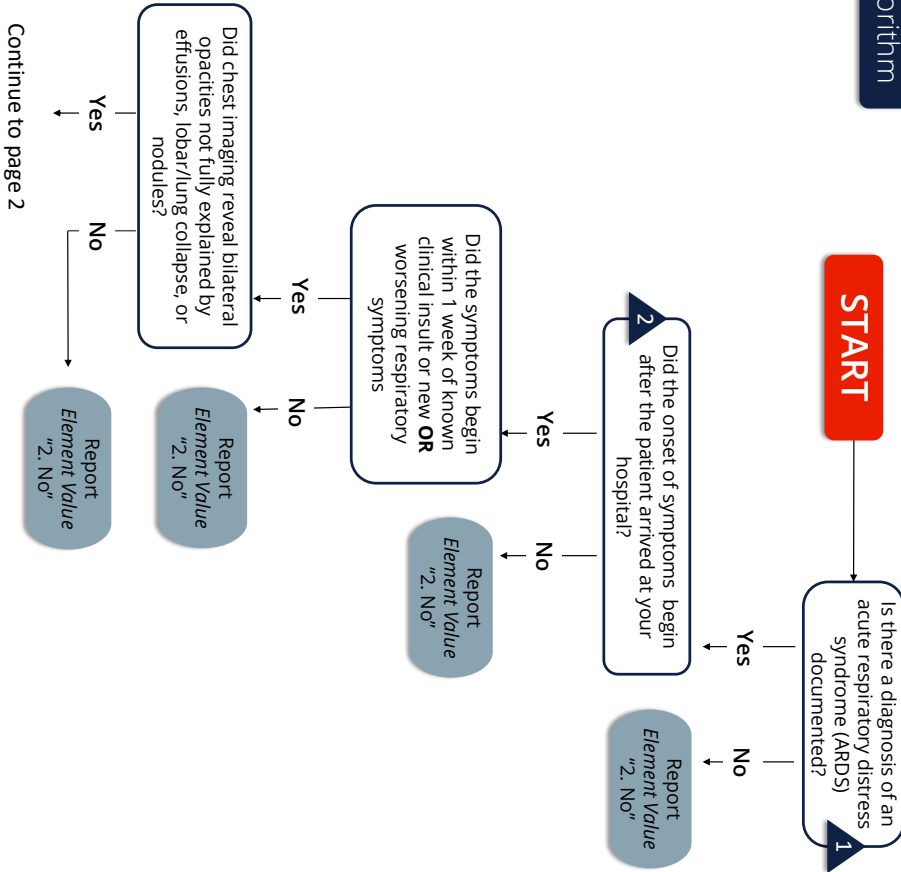
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3 Need objective assessment (e.g., echocardiography) to exclude hydrostatic edema if no risk factor present.

Acute Respiratory Distress Syndrome (ARDS) (pg. 1 of 2)

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1 The NTDS definition of ARDS is consistent with the 2012 New Berlin Definition.

2 Exclude if ARDS was present on arrival at your hospital.

ALCOHOL WITHDRAWAL SYNDROME

DESCRIPTION

Characterized by tremor, sweating, anxiety, agitation, depression, nausea, and malaise. It occurs 6-48 hours after cessation of alcohol consumption and, when uncomplicated, abates after 2-5 days. It may be complicated by grand mal seizures and may progress to delirium (known as delirium tremens).

ELEMENT VALUES

1. Yes
2. No

ADDITIONAL INFORMATION

- Onset of symptoms began after arrival to your ED/hospital.
- Documentation of alcohol withdrawal must be in the patient's medical record.
- Consistent with the 2019 World Health Organization (WHO) definition of Alcohol Withdrawal Syndrome.

DATA SOURCE HIERARCHY GUIDE

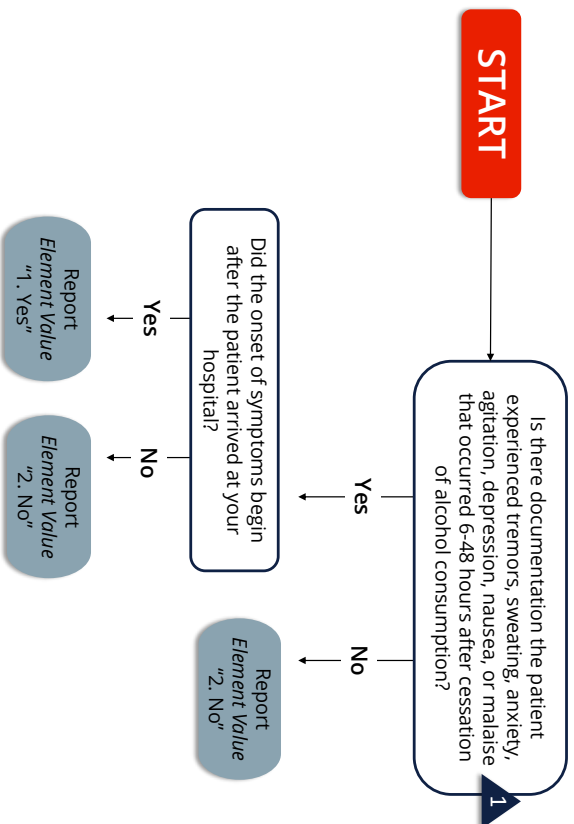
1. History and Physical
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ASSOCIATED EDIT CHECKS

Rule ID	Level	Message
18701	1	Value is not a valid menu option
18703	2	Element cannot be blank
18704	2	Element cannot be "Not Applicable"
18740	1	Single Entry Max exceeded

Alcohol Withdrawal Syndrome

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1 Note that a diagnosis of alcohol withdrawal syndrome is not required by the NTDS definition.

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Refer to complete NTDS definition for additional information

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CARDIAC ARREST WITH CPR

DESCRIPTION

Cardiac arrest is the sudden cessation of cardiac activity after hospital arrival. The patient becomes unresponsive with no normal breathing and no signs of circulation. If corrective measures are not taken rapidly, this condition progresses to sudden death.

INCLUDE:

- Patients who, after arrival at your hospital, have had an episode of cardiac arrest evaluated by hospital personnel, and received compressions or defibrillation or cardioversion or cardiac pacing to restore circulation.

EXCLUDE:

- Patients whose ONLY episode of cardiac arrest with CPR was on arrival to your hospital.

ELEMENT VALUES

1. Yes
2. No

ADDITIONAL INFORMATION

- Onset of symptoms began after arrival to your ED/hospital.
- Cardiac arrest must be documented in the patient's medical record.

DATA SOURCE HIERARCHY GUIDE

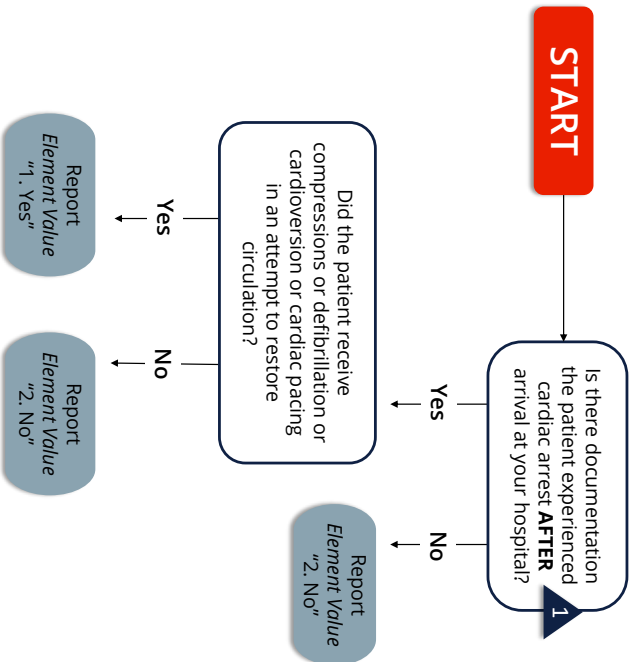
1. History and Physical
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ASSOCIATED EDIT CHECKS

Rule ID	Level	Message
18801	1	Value is not a valid menu option
18803	2	Element cannot be blank
18804	2	Element cannot be "Not Applicable"
18840	1	Single Entry Max exceeded

Cardiac Arrest with CPR

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1 Cardiac arrest is the sudden cessation of cardiac activity after hospital arrival. The patient becomes unresponsive with no normal breathing and no signs of circulation. If corrective measures are not taken rapidly, this condition progresses to sudden death.

EXCLUDE patients whose **ONLY** episode of cardiac arrest with CPR was on arrival to your hospital.

Consistent with the most recent version of the 2025 NTDS Data Dictionary
Refer to complete NTDS definition for additional information

CATHETER-ASSOCIATED URINARY TRACT INFECTION (CAUTI)

DESCRIPTION

A urinary tract infection (UTI) where an indwelling urinary catheter was in place for > 2 calendar days on the date of event, with day of device placement being Day 1,

AND

An indwelling urinary catheter was in place on the date of event or the day before. If an indwelling urinary catheter was in place for more than 2 consecutive days in an inpatient location and then removed, the date of event for the UTI must be the day of device discontinuation or the next day for the UTI to be catheter-associated.

January 2019 CDC CAUTI Criterion SUTI 1a:

Patient must meet 1, 2, **and** 3 below:

1. Patient had an indwelling urinary catheter that had been in place for more than 2 consecutive days in an inpatient location on the date of event AND was either:
 - Present for any portion of the calendar day on the date of event,

OR

 - Removed the day before the date of event.
2. Patient has at least one of the following signs or symptoms:
 - Fever (> 38°C): Reminder: To use fever in a patient > 65 years of age, the IUC needs to be in place for more than 2 consecutive days in an inpatient location on date of event and is either still in place or was removed the day before the DOE.
 - Suprapubic tenderness
 - Costovertebral angle pain or tenderness
 - Urinary urgency
 - Urinary frequency
 - Dysuria
3. Patient has a urine culture with no more than two species of organisms identified, at least one of which is a bacterium > 10⁵ CFU/ml.

January 2019 CDC CAUTI Criterion SUTI 2:

Patient must meet 1, 2 **and** 3 below:

1. Patient is ≤ 1 year of age
2. Patient has at least one of the following signs or symptoms:
 - fever (> 38.0°C)
 - hypothermia (< 36.0°C)
 - apnea
 - bradycardia
 - lethargy
 - vomiting
 - suprapubic tenderness
3. Patient has a urine culture with no more than two species of organisms, at least one of which is a bacterium of ≥ 10⁵ CFU/ml.

ELEMENT VALUES

1. Yes
2. No

ADDITIONAL INFORMATION

- Onset of symptoms began after arrival to your ED/hospital.
- A diagnosis of Urinary Tract Infection (UTI) must be documented in the patient's medical record.
- Consistent with the January 2019 CDC defined CAUTI.

DATA SOURCE HIERARCHY GUIDE

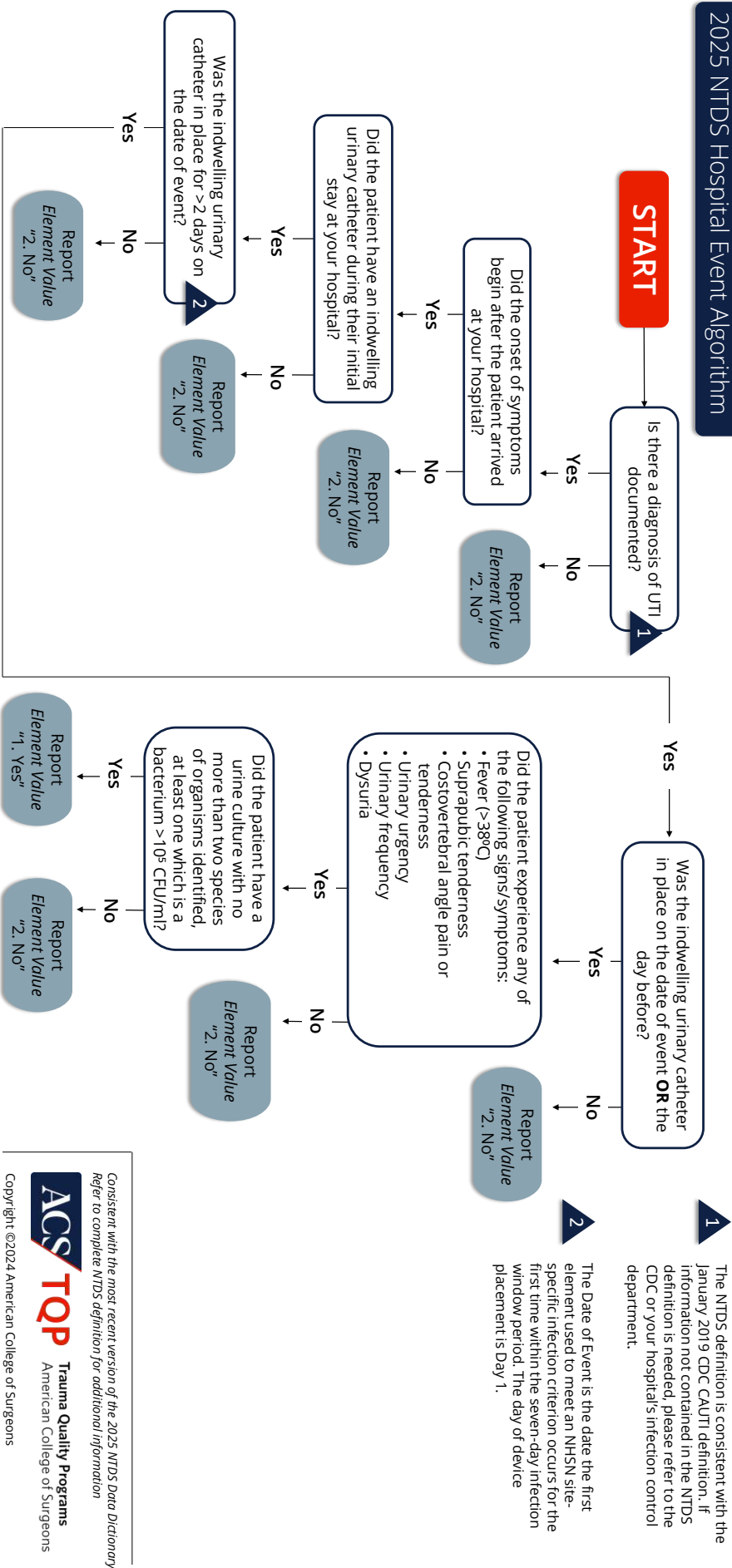
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ASSOCIATED EDIT CHECKS

Rule ID	Level	Message
18901	1	Value is not a valid menu option
18903	2	Element cannot be blank
18904	2	Element cannot be "Not Applicable"
18940	1	Single Entry Max exceeded

Catheter-associated Urinary Tract Infection (CAUTI) SUTI 1a (Patients >1 year of age)

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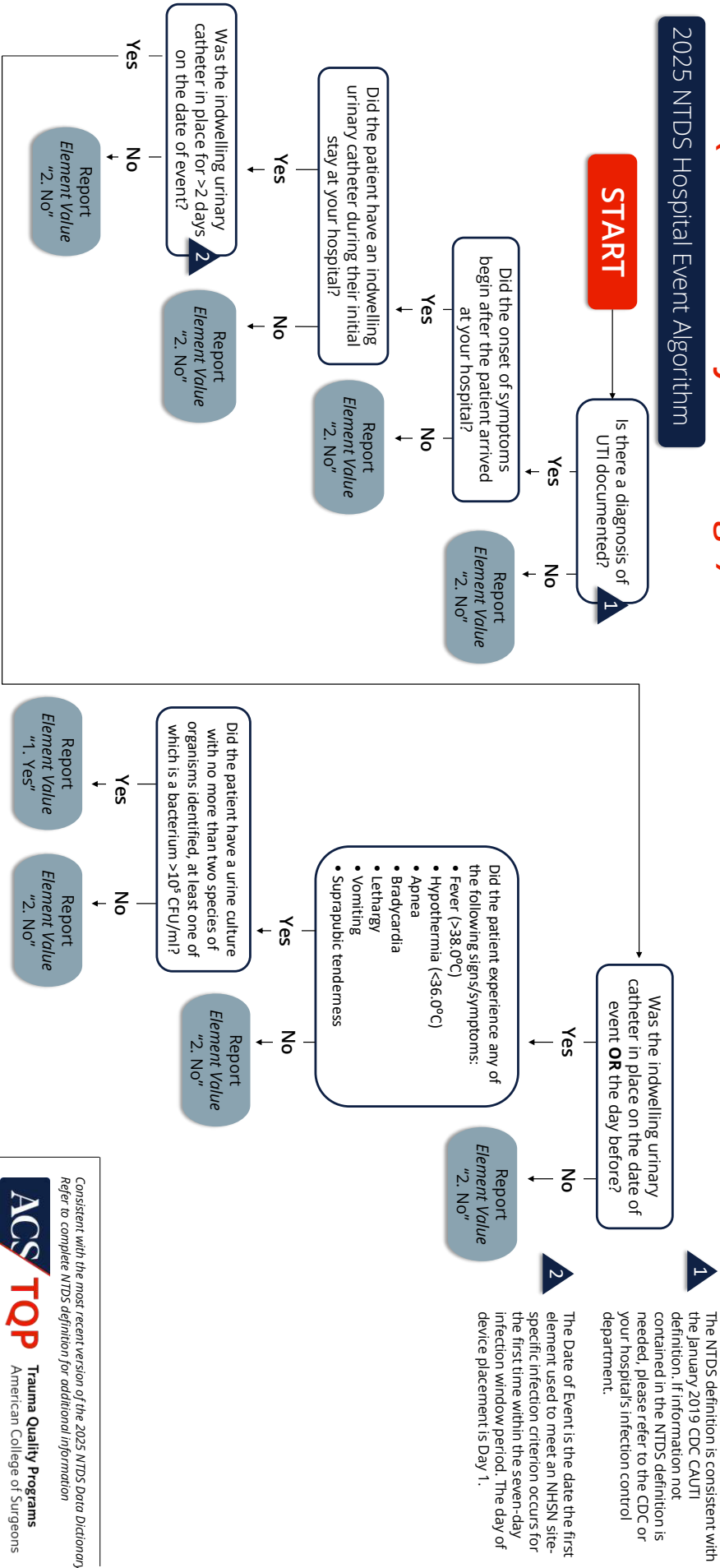
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Catheter-associated Urinary Tract Infection (CAUTI) SUTI 2 (Patients ≤ 1 year of age)

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CENTRAL LINE-ASSOCIATED BLOODSTREAM INFECTION (CLABSI)

DESCRIPTION

A laboratory-confirmed bloodstream infection (LCBI) where central line (CL) or umbilical catheter (UC) was in place for > 2 calendar days on the date of event, with day of device placement being Day 1,

AND

The line was also in place on the date of event or the day before. If a CL or UC was in place for > 2 calendar days and then removed, the date of event of the LCBI must be the day of discontinuation or the next day to be a CLABSI. If the patient is admitted or transferred into a facility with an implanted central line (port) in place, and that is the patient's only central line, day of first access in an inpatient location is considered Day 1. "Access" is defined as line placement, infusion or withdrawal through the line. Such lines continue to be eligible for CLABSI once they are accessed until they are either discontinued or the day after patient discharge (as per the Transfer Rule). Note that the "de-access" of a port does not result in the patient's removal from CLABSI surveillance.

January 2016 CDC Criterion LCBI 1:

Patient has a recognized pathogen identified from one or more blood specimens by a culture or non-culture based microbiologic testing method which is performed for purposes of clinical diagnosis or treatment (e.g., not Active Surveillance Culture/Testing (ASC/AST).

AND

Organism(s) identified in blood is not related to an infection at another site.

OR

January 2016 CDC Criterion LCBI 2:

Patient has at least one of the following signs or symptoms: fever (>38°C), chills, or hypotension.

AND

Organism(s) identified from blood is not related to an infection at another site.

AND

The same common commensal (i.e., diphtheroids [*Corynebacterium* spp. not *C. diphtheriae*], *Bacillus* spp. [not *B. anthracis*], *Propionibacterium* spp., coagulase-negative staphylococci [including *S. epidermidis*], viridans group streptococci, *Aerococcus* spp., and *Micrococcus* spp.) is identified from two or more blood specimens drawn on separate occasions, by a culture or non-culture based microbiologic testing method which is performed for purposes of clinical diagnosis or treatment (e.g., not Active Surveillance Culture/Testing (ASC/AST). Criterion elements must occur within the Infection Window Period, the 7-day time period which includes the collection date of the positive blood, the 3 calendar days before and the 3 calendar days after.

OR

January 2016 CDC Criterion LCBI 3:

Patient ≤ 1 year of age has at least one of the following signs or symptoms: fever ($>38^{\circ}\text{C}$), hypothermia ($<36^{\circ}\text{C}$), apnea, or bradycardia.

AND

Organism(s) identified from blood is not related to an infection at another site.

AND

The same common commensal (i.e., diphtheroids [*Corynebacterium* spp. not *C. diphtheriae*], *Bacillus* spp. [not *B. anthracis*], *Propionibacterium* spp., coagulase-negative staphylococci [including *S. epidermidis*], viridans group streptococci, *Aerococcus* spp., *Micrococcus* spp.) is identified from two or more blood specimens drawn on separate occasions, by a culture or non-culture base microbiologic testing method which is performed for purposes of clinical diagnosis or treatment (e.g., not Active Surveillance Culture/Testing (ASC/AST). Criterion elements must occur within the Infection Window Period, the 7-day time period which includes the collection date of the positive blood, the 3 calendar days before and the 3 calendar days after.

ELEMENT VALUES

1. Yes
2. No

ADDITIONAL INFORMATION

- Onset of symptoms began after arrival to your ED/hospital.
- A diagnosis of central line-associated bloodstream infection (CLABSI) must be documented in the patient's medical record.
- Consistent with the January 2016 CDC defined CLABSI.

DATA SOURCE HIERARCHY GUIDE

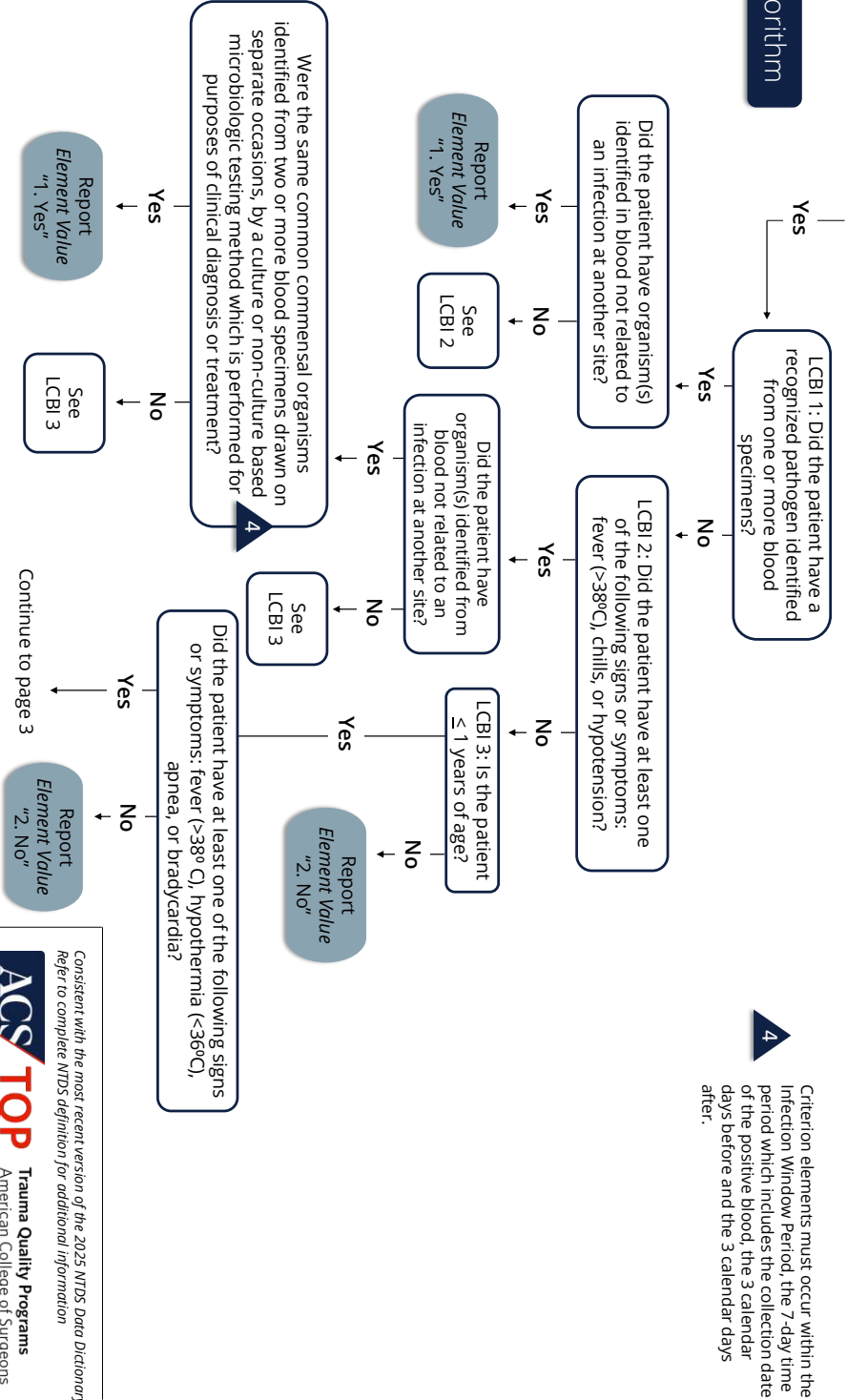
1. History and Physical
2. Physician Notes/Flow Sheet
3. Progress Notes
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7. Discharge Summary

ASSOCIATED EDIT CHECKS

Rule ID	Level	Message
19001	1	Value is not a valid menu option
19003	2	Element cannot be blank
19004	2	Element cannot be "Not Applicable"
19040	1	Single Entry Max exceeded

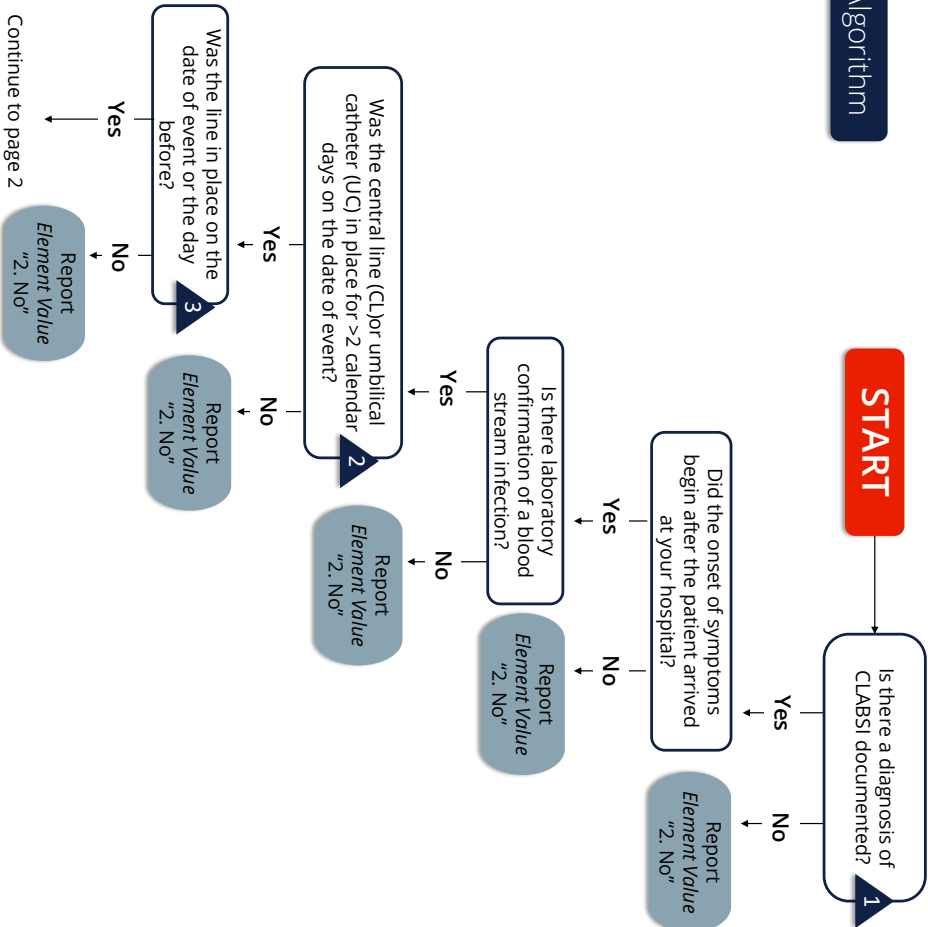
Central Line-Associated Blood Stream Infection (CLABSI) (pg. 2 of 3)

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Central Line-Associated Blood Stream Infection (CLABSI) (pg. 1 of 3)

2025 NTDS Hospital Event Algorithm



1 The NTDS definition is consistent with the January 2016 CDC CLABSI definition. If information not contained in the NTDS definition is needed, please refer to the CDC or your hospital's infection control department.

2 The Date of Event is the date the first element used to meet an NHSN site-specific infection criterion occurs for the first time within the seven-day infection window period. The day of device placement is Day 1.

3 If a CL or UC was in place for > 2 calendar days and then removed, the date of event of the LCBI must be the day of discontinuation or the next day to be a CLABSI. If the patient is admitted or transferred into a facility with an implanted central line (port) in place, and that is the patient's only central line, day of first access in an inpatient location is considered Day 1. "Access" is defined as line placement, infusion or withdrawal through the line. Such lines continue to be eligible for CLABSI once they are accessed until they are either discontinued or the day after patient discharge (as per the Transfer Rule). Note that the "de-access" of a port does not result in the patient's removal from CLABSI surveillance.

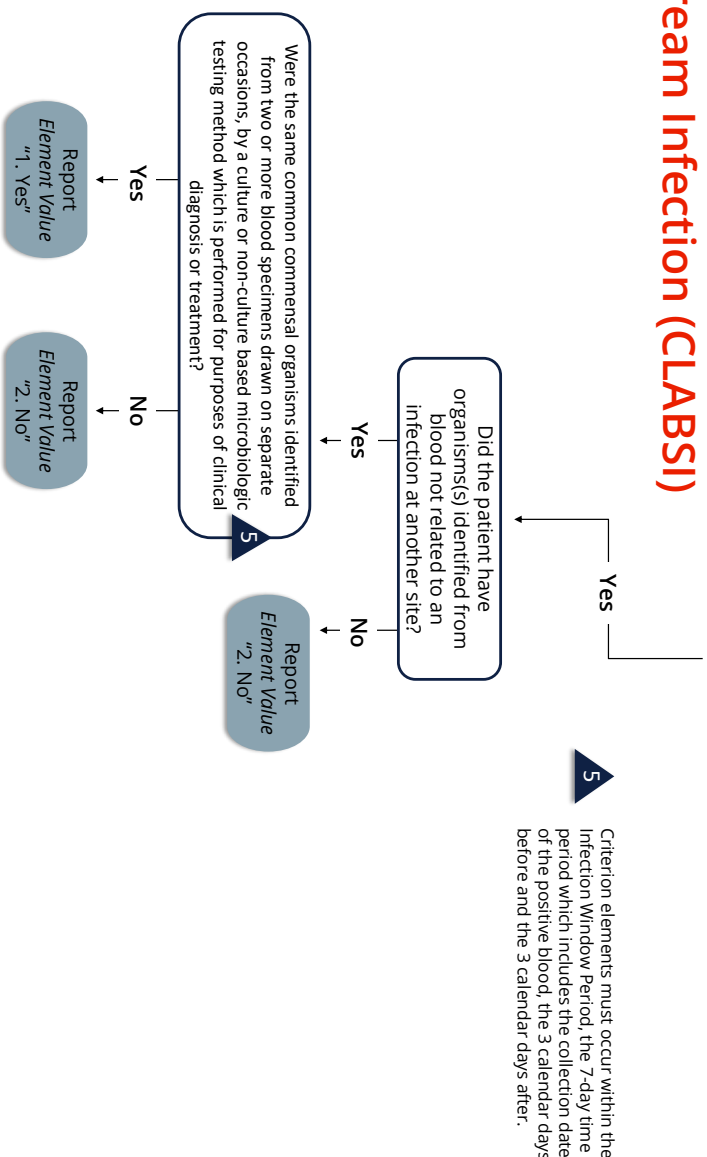
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Central Line-Associated Blood Stream Infection (CLABSI) (pg. 3 of 3)

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DEEP SURGICAL SITE INFECTION

DESCRIPTION

Must meet the following criteria:

Infection occurs within 30 or 90 days after the NHSN operative procedure (where day 1 = the procedure date) according to list in Table 2

AND

Involves deep soft tissues of the incision (e.g., fascial and muscle layers)

AND

Patient has at least *one* of the following:

- a. Purulent drainage from the deep incision.
- b. A deep incision that spontaneously dehisces, or is deliberately opened or aspirated by a surgeon, attending physician** or other designee and organism is identified by a culture or non-culture based microbiologic testing method which is performed for purposes of clinical diagnosis or treatment (e.g., not Active Surveillance Culture/Testing (ASC/AST) or culture or non-culture based microbiologic testing method is not performed.

AND

Organism(s) identified from the deep soft tissues of the incision by culture or non-culture based microbiologic testing method which is performed for purposes of clinical diagnosis or treatment (for example, not Active Surveillance Culture/Testing (ASC/AST) or culture or non-culture based microbiologic testing method is not performed. A culture or non-culture based test from the deep soft tissues of the incision that has a negative finding does not meet this criterion.

AND

Patient has at least one of the following signs or symptoms: fever ($> 38^{\circ}\text{C}$); localized pain or tenderness. A culture or non-culture based test that has a negative finding does not meet this criterion.

- c. An abscess or other evidence of infection involving the deep incision that is detected on gross anatomical or histopathologic exam, or imaging test

* The term attending physician for the purposes of application of the NHSN SSI criteria may be interpreted to mean the surgeon(s), infectious disease, other physician on the case, emergency physician, or physician's designee (nurse practitioner or physician's assistant).

COMMENTS: There are two specific types of deep incisional SSIs:

1. Deep Incisional Primary (DIP) – a deep incisional SSI that is identified in a primary incision in a patient that has had an operation with one or more incisions (e.g., C-section incision or chest incision for CBGB).
2. Deep Incisional Secondary (DIS) – a deep incisional SSI that is identified in the secondary incision in a patient that has had an operation with more than one incision (e.g., donor site incision for CBGB).

Table 2. Surveillance Period for Deep Incisional or Organ/Space SSI Following Selected NHSN Operative Procedure Categories. Day 1 = the date of the procedure.

30 DAY SURVEILLANCE			
Code	Operative Procedure	Code	Operative Procedure
AAA	Abdominal aortic aneurysm repair	LAM	Laminectomy
AMP	Limb amputation	LTP	Liver transplant
APPY	Appendix surgery	NECK	Neck surgery
AVSD	Shunt for dialysis	NEPH	Kidney surgery
BILI	Bile duct, liver or pancreatic surgery	OVRY	Ovarian surgery
CEA	Carotid endarterectomy	PRST	Prostate surgery
CHOL	Gallbladder surgery	REC	Rectal surgery
COLO	Colon surgery	SB	Small bowel surgery
CSEC	Cesarean section	SPLE	Spleen surgery
GAST	Gastric surgery	THOR	Thoracic surgery
HTP	Heart transplant	THUR	Thyroid and/or parathyroid surgery
HYST	Abdominal hysterectomy	VHYS	Vaginal hysterectomy
KTP	Kidney transplant	XLAP	Exploratory Laparotomy

90 DAY SURVEILLANCE	
Code	Operative Procedure
BRST	Breast surgery
CARD	Cardiac surgery
CBGB	Coronary artery bypass graft with both chest and donor site incisions
CBGC	Coronary artery bypass graft with chest incision only
CRAN	Craniotomy
FUSN	Spinal fusion
FX	Open reduction of fracture
HER	Herniorrhaphy
HPRO	Hip prosthesis
KPRO	Knee prosthesis
PACE	Pacemaker surgery
PVBY	Peripheral vascular bypass surgery
VSHN	Ventricular shunt

ELEMENT VALUES

1. Yes
2. No

ADDITIONAL INFORMATION

- Onset of symptoms began after arrival to your ED/hospital.
- A diagnosis of a surgical site infection must be documented in the patient's medical record.
- Consistent with the CDC January 2019 defined SSI.

DATA SOURCE HIERARCHY GUIDE

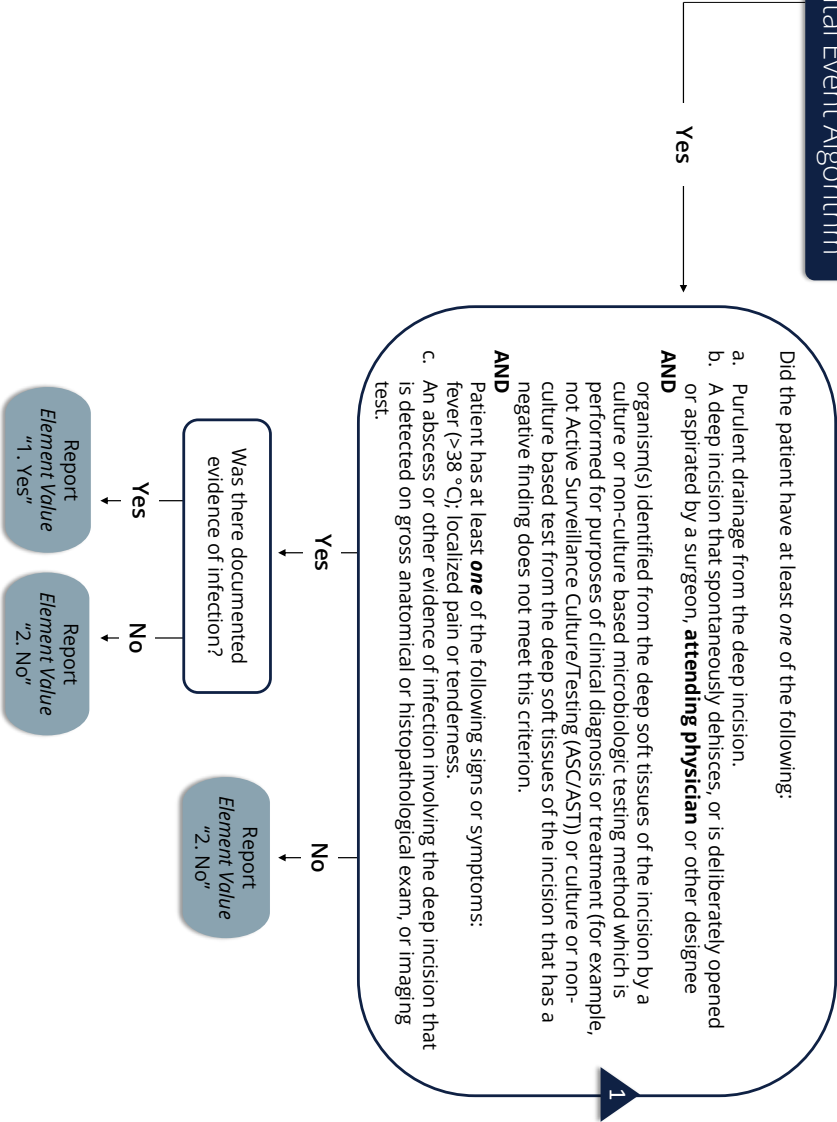
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ASSOCIATED EDIT CHECKS

Rule ID	Level	Message
19101	1	Value is not a valid menu option
19103	2	Element cannot be blank
19104	2	Element cannot be "Not Applicable"
19140	1	Single Entry Max exceeded

Deep Surgical Site Infection (Deep SSI) (pg. 2 of 2)

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1

The term **attending physician** for the purposes of application of the NHSN SSI criteria may be interpreted to mean the surgeon(s), infectious disease, other physician on the case, emergency physician, or physician's designee (nurse practitioner or physician's assistant).

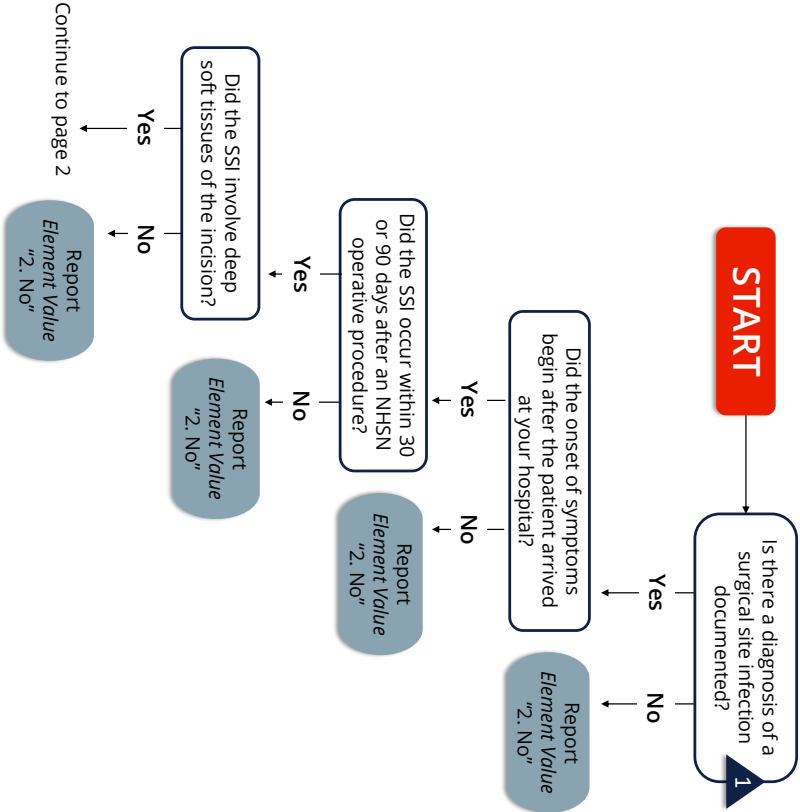
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Deep Surgical Site Infection (Deep SSI) (pg. 1 of 2)

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1

The NTDS definition is consistent with the January 2019 CDC SSI definition. If information not contained in the NTDS definition is needed, please refer to the CDC or your hospital's infection control department.

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DEEP VEIN THROMBOSIS (DVT)

DESCRIPTION

The formation, development, or existence of a blood clot or thrombus within the venous system, which may be coupled with inflammation.

ELEMENT VALUES

1. Yes
2. No

ADDITIONAL INFORMATION

- Onset of symptoms began after arrival to your ED/hospital.
- A diagnosis of deep vein thrombosis (DVT) must be documented in the patient's medical record, which may be confirmed by venogram, ultrasound, or CT.
- The patient must be treated with anticoagulation therapy and/or placement of a vena cava filter or clipping of the vena cava.

DATA SOURCE HIERARCHY GUIDE

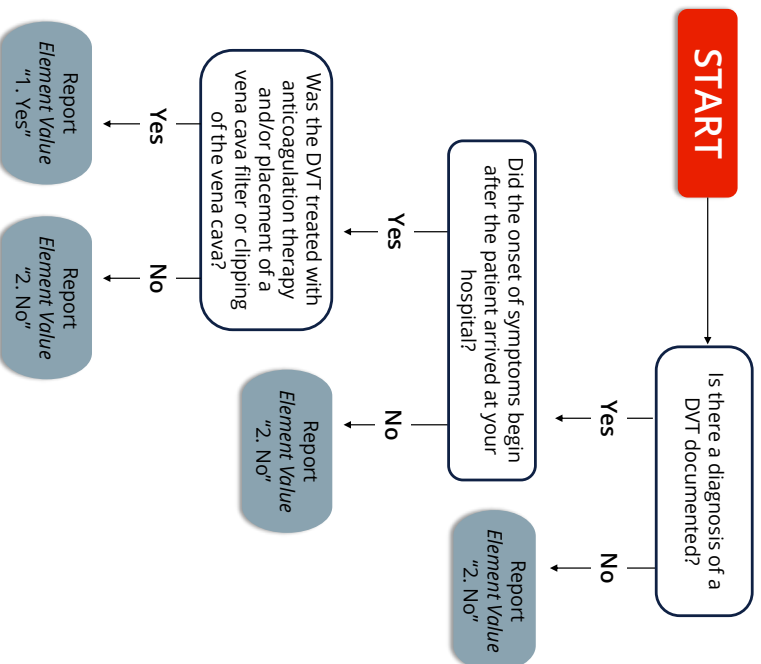
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ASSOCIATED EDIT CHECKS

Rule ID	Level	Message
19201	1	Value is not a valid menu option
19203	2	Element cannot be blank
19204	2	Element cannot be "Not Applicable"
19240	1	Single Entry Max exceeded

Deep Vein Thrombosis (DVT)

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DELIRIUM

DESCRIPTION

Acute onset of behaviors characterized by restlessness, illusions, and incoherence of thought and speech. Delirium can often be traced to one or more contributing factors, such as a severe or chronic medical illness, changes in your metabolic balance (such as low sodium), medication, infection, surgery, or alcohol or drug withdrawal.

OR

Patient tests positive after using an objective screening tool like the Confusion Assessment Method (CAM) or the Intensive Care Delirium Screening Checklist (ICDSC).

OR

A diagnosis of delirium documented in the patient's medical record.

EXCLUDE:

- Patients whose delirium is due to alcohol withdrawal.

ELEMENT VALUES

1. Yes
2. No

ADDITIONAL INFORMATION

- Onset of symptoms began after arrival to your ED/hospital.

DATA SOURCE HIERARCHY GUIDE

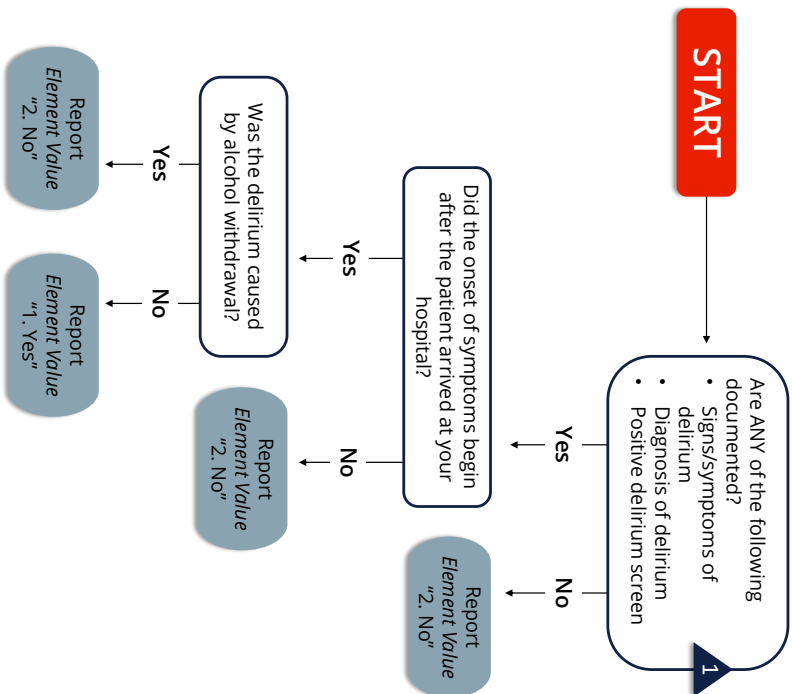
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ASSOCIATED EDIT CHECKS

Rule ID	Level	Message
21601	1	Value is not a valid menu option
21603	2	Element cannot be blank
21604	2	Element cannot be "Not Applicable"
21640	1	Single Entry Max exceeded

Delirium

2025 NTDS Hospital Event Algorithm



1

Delirium can often be traced to one or more contributing factors, such as a severe or chronic medical illness, changes in your metabolic balance (such as low sodium), medication, infection, surgery, or alcohol or drug withdrawal.

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MYOCARDIAL INFARCTION (MI)

DESCRIPTION

An acute myocardial infarction (MI) must be noted with documentation of ECG changes indicative of an acute MI

AND

New elevation in troponin greater than three times upper level of the reference range in the setting of suspected myocardial ischemia

AND

Physician diagnosis of an acute myocardial infarction that occurred subsequent to arrival at your hospital.

ELEMENT VALUES

1. Yes
2. No

ADDITIONAL INFORMATION

- Onset of symptoms began after arrival to your ED/hospital.

DATA SOURCE HIERARCHY GUIDE

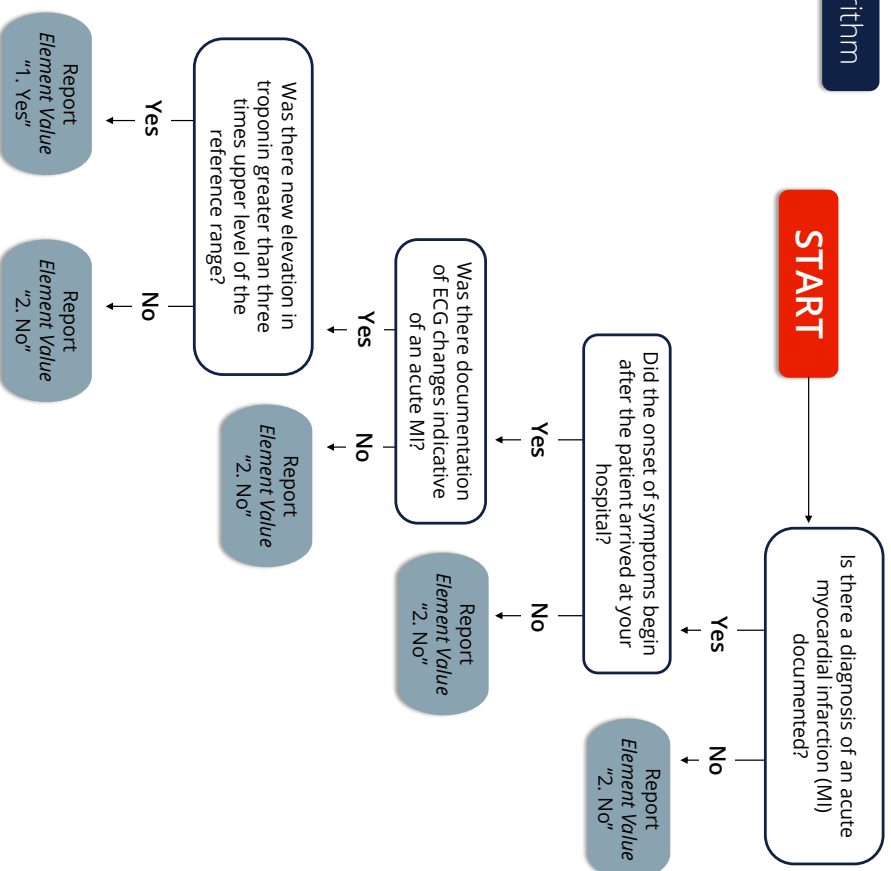
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ASSOCIATED EDIT CHECKS

Rule ID	Level	Message
19401	1	Value is not a valid menu option
19403	2	Element cannot be blank
19404	2	Element cannot be "Not Applicable"
19440	1	Single Entry Max exceeded

Myocardial Infarction

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ORGAN/SPACE SURGICAL SITE INFECTION

DESCRIPTION

Must meet the following criteria:

Infection occurs within 30 or 90 days after the NHSN operative procedure (where day 1 = the procedure date) according to the list in Table 2

AND

Infection involves any part of the body deeper than the fascial/muscle layers, that is opened or manipulated during the operative procedure

AND

Patient has at least *one* of the following:

- Purulent drainage from a drain that is placed into the organ/space (e.g., closed suction drainage system, open drain, T-tube drain, CT guided drainage).
- Organisms are identified from fluid or tissue in the organ/space by a culture or non-culture based microbiologic testing method which is performed for purposes of clinical diagnosis or treatment (e.g., not Active Surveillance Culture/Testing (ASC/AST)).
- An abscess or other evidence of infection involving the organ/space that is detected on gross anatomical or histopathologic exam, or imaging test evidence suggestive of infection.

AND

Meets at least one criterion for a specific organ/space infection site listed in Table 3. These criteria are found in the Surveillance Definitions for Specific Types of Infections chapter.

Table 2. Surveillance Period for Deep Incisional or Organ/Space SSI Following Selected NHSN Operative Procedure Categories. Day 1 = the date of the procedure.

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CEA	Carotid endarterectomy	PRST	Prostate surgery
CHOL	Gallbladder surgery	REC	Rectal surgery
COLO	Colon surgery	SB	Small bowel surgery
CSEC	Cesarean section	SPLE	Spleen surgery
GAST	Gastric surgery	THOR	Thoracic surgery
HTP	Heart transplant	THUR	Thyroid and/or parathyroid surgery
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KTP	Kidney transplant	XLAP	Exploratory Laparotomy

90 DAY SURVEILLANCE	
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CARD	Cardiac surgery
CBGB	Coronary artery bypass graft with both chest and donor site incisions
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CRAN	Craniotomy
FUSN	Spinal fusion
FX	Open reduction of fracture
HER	Herniorrhaphy
HPRO	Hip prosthesis
KPRO	Knee prosthesis
PACE	Pacemaker surgery
PVBY	Peripheral vascular bypass surgery
VSHN	Ventricular shunt

Table 3. Specific Sites of an Organ/Space SSI.

Code	SITE	Code	SITE
BONE	Osteomyelitis	MED	Mediastinitis
BRST	Breast abscess or mastitis	MEN	Meningitis or ventriculitis
CARD	Myocarditis or pericarditis	ORAL	Oral cavity infection (mouth, tongue, or gums)
DISC	Disc space infection	OREP	Deep pelvic tissue infection or other infection of the male or female reproductive tract
EAR	Ear, mastoid infection	PJI	Periprosthetic Joint Infection
EMET	Endometritis	SA	Spinal abscess/infection
ENDO	Endocarditis	SINU	Sinusitis
GIT	Gastrointestinal (GI) tract infection	UR	Upper respiratory tract, pharyngitis, laryngitis, epiglottitis
IAB	Intraabdominal infection, not specified elsewhere	USI	Urinary System Infection
IC	Intracranial infection	VASC	Arterial or venous infection
JNT	Joint or bursa infection	VCUF	Vaginal cuff infection
LUNG	Other infection of the lower respiratory tract		

ELEMENT VALUES

1. Yes

2. No

ADDITIONAL INFORMATION

- Onset of symptoms began after arrival to your ED/hospital.
- A diagnosis of a surgical site infection must be documented in the patient's medical record.
- Consistent with the CDC January 2019 defined SSI.

DATA SOURCE HIERARCHY GUIDE

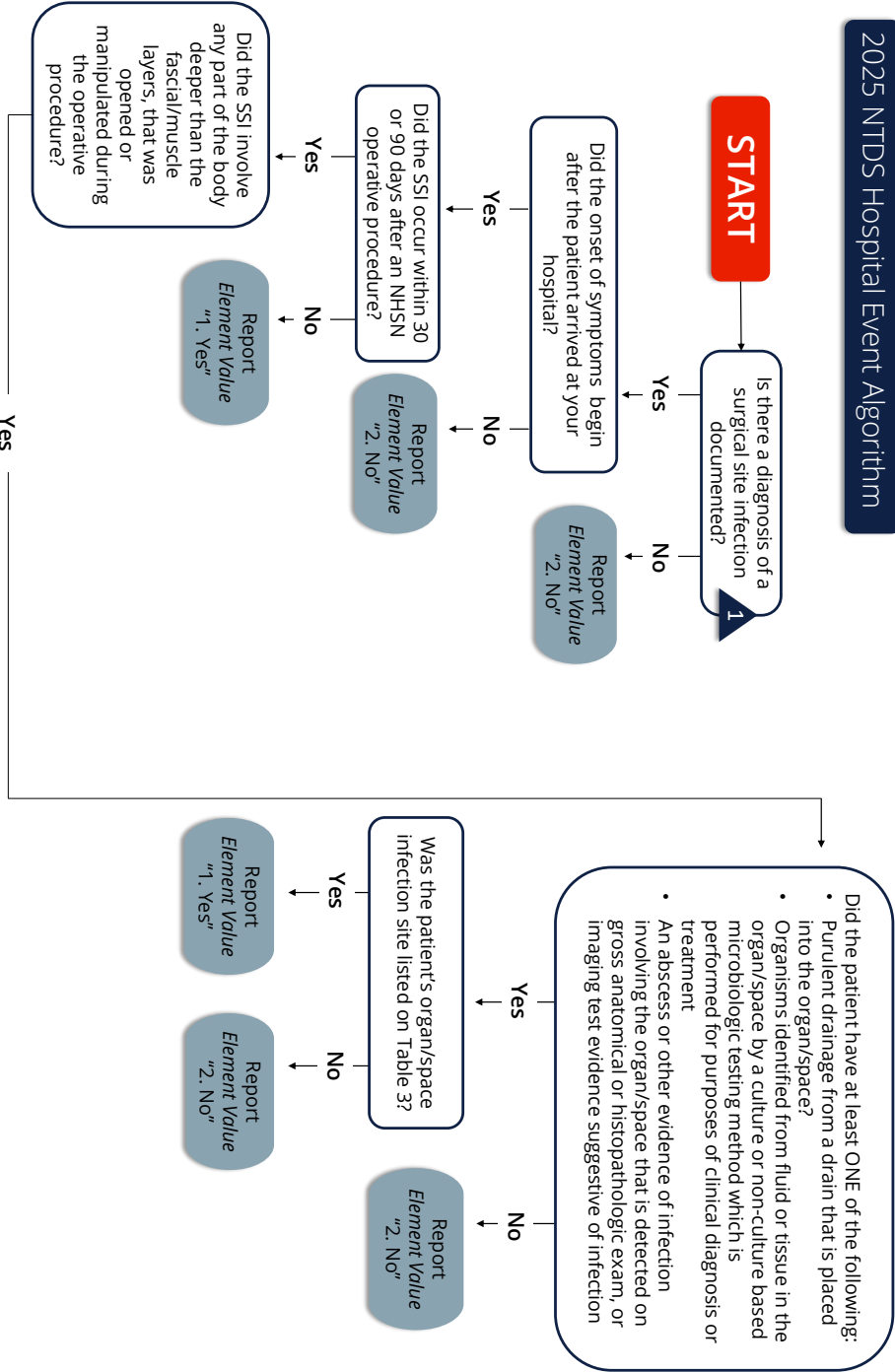
1. History and Physical
2. Physician Notes/Flow Sheet
3. Progress Notes
4. Case Management/Social Services Notes
5. Nursing Notes/Flow Sheet
6. Triage/Trauma Flow Sheet
7. Discharge Summary

ASSOCIATED EDIT CHECKS

Rule ID	Level	Message
19501	1	Value is not a valid menu option
19503	2	Element cannot be blank
19504	2	Element cannot be "Not Applicable"
19540	1	Single Entry Max exceeded

Organ/Space Surgical Site Infection (O/S SSI)

2025 NTDS Hospital Event Algorithm



1

The NTDS definition is consistent with the January 2019 CDC SSI definition. If information not contained in the NTDS definition is needed, please refer to the CDC or your hospital's infection control department.

Consistent with the most recent version of the 2025 NTDS Data Dictionary
Refer to complete NTDS definition for additional information

ACS TQP Trauma Quality Programs
American College of Surgeons
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OSTEOMYELITIS

DESCRIPTION

Osteomyelitis must meet at least one of the following criteria:

1. Patient has organism(s) identified from bone by culture or non-culture based microbiologic testing method, which is performed for purposes of clinical diagnosis and treatment, for example, not Active Surveillance Culture/Testing (ASC/AST).
2. Patient has evidence of osteomyelitis on gross anatomic or histopathologic exam.
3. Patient has at least two of the following localized signs or symptoms:
 - Fever ($> 38.0^{\circ}\text{C}$)
 - Swelling*
 - Pain or tenderness*
 - Heat*
 - Drainage*

AND at least one of the following:

- a. Organisms identified from blood by culture or non-culture based microbiologic testing method, which is performed for purposes of clinical diagnosis and treatment, for example, not Active Surveillance Culture/Testing (ASC/AST) AND Imaging test evidence suggestive of infection (for example, x-ray, CT scan, MRI, radiolabel scan [gallium, technetium, etc.]), which if equivocal is supported by clinical correlation, specifically, physician documentation of antimicrobial treatment for osteomyelitis.
- b. Imaging test evidence suggestive of infection (for example, x-ray, CT scan, MRI, radiolabel scan [gallium, technetium, etc.]), which if equivocal is supported by clinical correlation, specifically, physician documentation of antimicrobial treatment for osteomyelitis).

*With no other recognized cause

ELEMENT VALUES

1. Yes
2. No

ADDITIONAL INFORMATION

- Onset of symptoms began after arrival to your ED/hospital.
- A diagnosis of osteomyelitis must be documented in the patient's medical record.
- Consistent with the January 2020 CDC definition of Bone and Joint Infection.

DATA SOURCE HIERARCHY GUIDE

1. History and Physical
2. Physician Notes/Flow Sheet
3. Progress Notes
4. Case Management/Social Services Notes
5. Nursing Notes/Flow Sheet
6. Triage/Trauma Flow Sheet
7. Discharge Summary

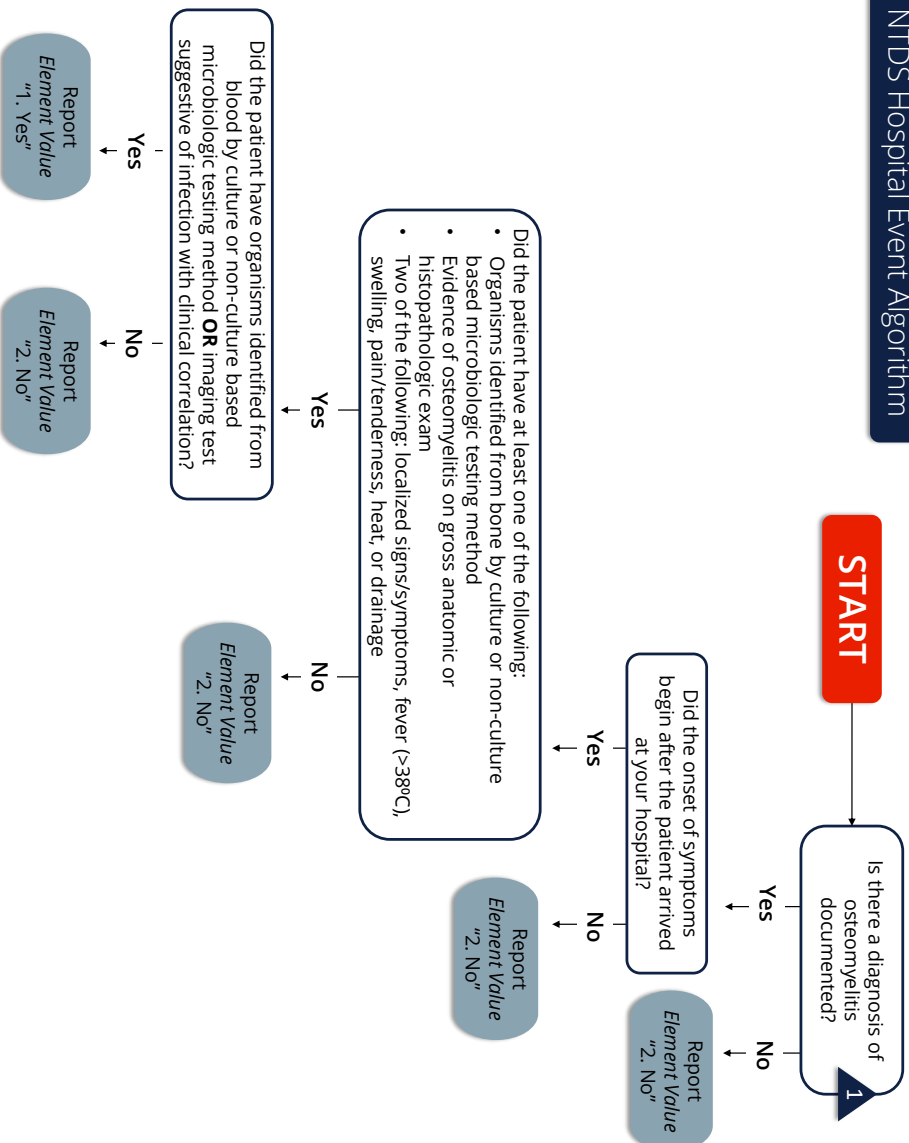
ASSOCIATED EDIT CHECKS

Rule ID	Level	Message
19601	1	Value is not a valid menu option
19603	2	Element cannot be blank
19604	2	Element cannot be "Not Applicable"
19640	1	Single Entry Max exceeded

HOSPITAL EVENTS

Osteomyelitis

2025 NTDS Hospital Event Algorithm



1

The NTDS definition is consistent with the January 2020 CDC Bone and Joint Infection definition. If information not contained in the NTDS definition is needed, please refer to the CDC or your hospital's infection control department.

PRESSURE ULCER

DESCRIPTION

A localized injury to the skin and/or underlying tissue usually over a bony prominence, as a result of pressure, or pressure in combination with shear. A number of contributing or confounding factors are also associated with pressure ulcers; the significance of these factors is yet to be elucidated. Equivalent to NPUAP Stages II-IV, Unstageable/Unclassified, and Suspected Deep Tissue Injury.

ELEMENT VALUES

1. Yes
2. No

ADDITIONAL INFORMATION

- Onset of NPUAP Stage II began after arrival to your ED/hospital.
- Pressure ulcer documentation must be in the patient's medical record.
- Consistent with the NPUAP 2014.

DATA SOURCE HIERARCHY GUIDE

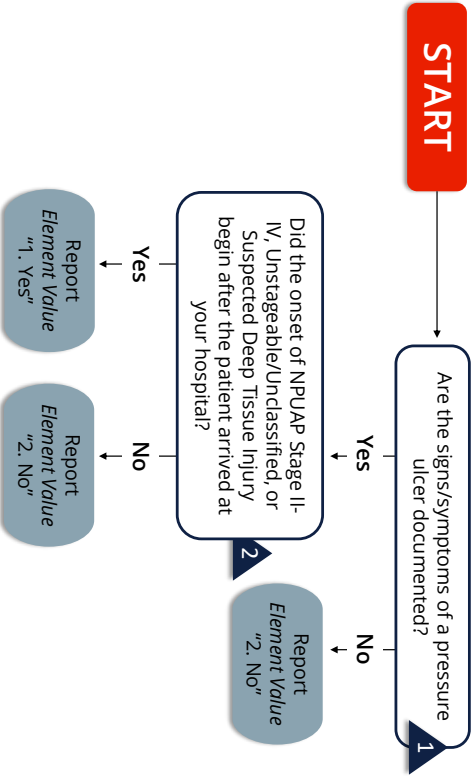
1. History and Physical
2. Physician Notes/Flow Sheet
3. Progress Notes
4. Case Management/Social Services Notes
5. Nursing Notes/Flow Sheet
6. Triage/Trauma Flow Sheet
7. Discharge Summary

ASSOCIATED EDIT CHECKS

Rule ID	Level	Message
19801	1	Value is not a valid menu option
19803	2	Element cannot be blank
19804	2	Element cannot be "Not Applicable"
19840	1	Single Entry Max exceeded

Pressure Ulcer

2025 NTDS Hospital Event Algorithm



1

A pressure ulcer is a localized injury to the skin and/or underlying tissue usually over a bony prominence, as a result of pressure, or pressure in combination with shear. A number of contributing or confounding factors are also associated with pressure ulcers; the significance of these factors is yet to be elucidated.

2

The NTDS definition is consistent with the NPUAP 2014.

Consistent with the most recent version of the 2025 NTDS Data Dictionary
Refer to complete NTDS definition for additional information



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PULMONARY EMBOLISM (PE)

DESCRIPTION

A lodging of a blood clot in a pulmonary artery with subsequent obstruction of blood supply to the lung parenchyma. The blood clots usually originate from the deep leg veins or the pelvic venous system.

EXCLUDE:

- Subsegmental PEs.

ELEMENT VALUES

1. Yes
2. No

ADDITIONAL INFORMATION

- Onset of symptoms began after arrival to your ED/hospital.
- Consider the condition present if the patient has a VQ scan interpreted as high probability of pulmonary embolism or a positive pulmonary arteriogram or positive CT angiogram and/or a diagnosis of PE is documented in the patient's medical record.

DATA SOURCE HIERARCHY GUIDE

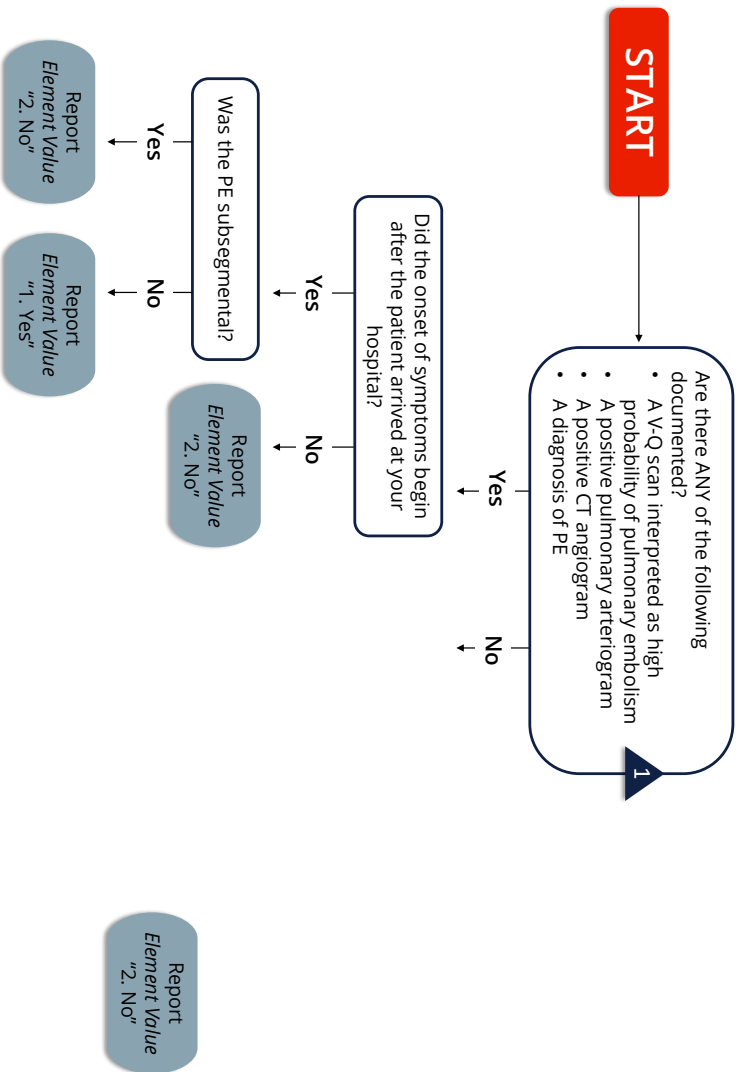
1. History and Physical
2. Physician Notes/Flow Sheet
3. Progress Notes
4. Case Management/Social Services Notes
5. Nursing Notes/Flow Sheet
6. Triage/Trauma Flow Sheet
7. Discharge Summary

ASSOCIATED EDIT CHECKS

Rule ID	Level	Message
19701	1	Value is not a valid menu option
19703	2	Element cannot be blank
19704	2	Element cannot be "Not Applicable"
19740	1	Single Entry Max exceeded

Pulmonary Embolism (PE)

2025 NTDS Hospital Event Algorithm



1 Subsegmental PEs are excluded from the NTDS definition of PE.

Consistent with the most recent version of the 2025 NTDS Data Dictionary
Refer to complete NTDS definition for additional information



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SEVERE SEPSIS

DESCRIPTION

Severe sepsis: sepsis plus organ dysfunction, hypotension (low blood pressure), or hypoperfusion (insufficient blood flow) to 1 or more organs.

Septic shock: sepsis with persisting arterial hypotension or hypoperfusion despite adequate fluid resuscitation.

ELEMENT VALUES

1. Yes
2. No

ADDITIONAL INFORMATION

- Onset of symptoms began after arrival to your ED/hospital.
- A diagnosis of sepsis must be documented in the patient's medical record.
- Consistent with the American College of Chest Physicians and the Society of Critical Care Medicine October 2010.

DATA SOURCE HIERARCHY GUIDE

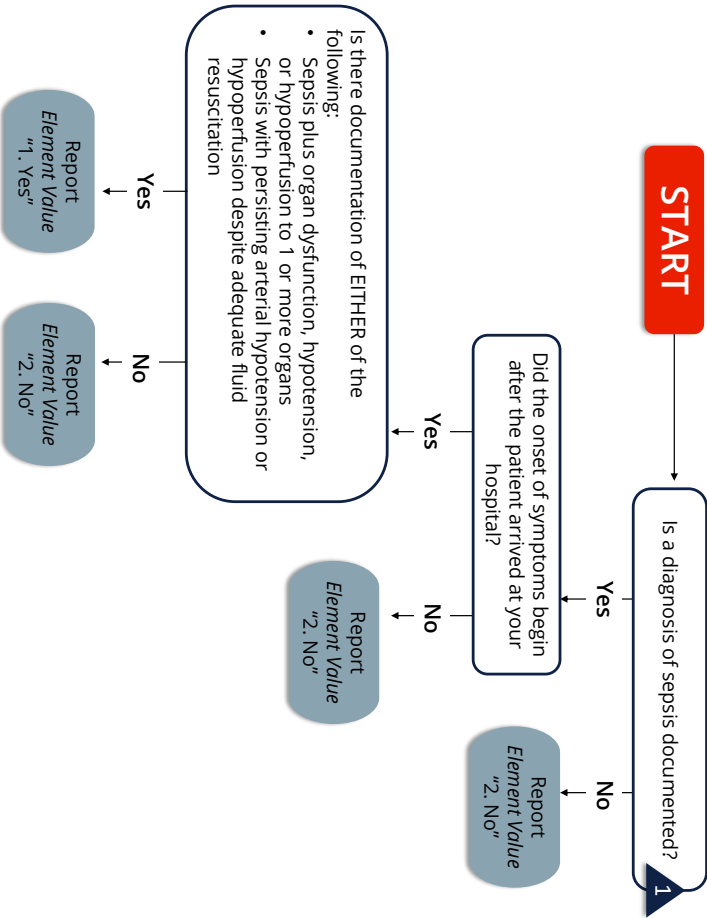
1. History and Physical
2. Physician Notes/Flow Sheet
3. Progress Notes
4. Case Management/Social Services Notes
5. Nursing Notes/Flow Sheet
6. Triage/Trauma Flow Sheet
7. Discharge Summary

ASSOCIATED EDIT CHECKS

Rule ID	Level	Message
19901	1	Value is not a valid menu option
19903	2	Element cannot be blank
19904	2	Element cannot be "Not Applicable"
19940	1	Single Entry Max exceeded

Severe Sepsis

2025 NTDS Hospital Event Algorithm



1 The NTDS definition is consistent with the American College of Chest Physicians and the Society of Critical Care Medicine October 2010.

DESCRIPTION

A focal or global neurological deficit of rapid onset and NOT present on admission caused by a clot obstructing the flow of blood flow to the brain (ischemic stroke). Or by a blood vessel rupturing and preventing blood flow to the brain (hemorrhagic stroke). Or a transient ischemic attack which is temporary caused by a temporary clot. The patient must have at least one of the following symptoms:

- Change in level of consciousness
- Hemiplegia
- Hemiparesis
- Numbness or sensory loss affecting on side of the body
- Dysphasia or aphasia
- Hemianopia
- Amaurosis fugax
- Other neurological signs or symptoms consistent with stroke

AND

- Duration of neurological deficit ≥ 24 h

OR

- Duration of deficit < 24 h, if neuroimaging (MR, CT, or cerebral angiography) documents a new hemorrhage or infarct consistent with stroke, or therapeutic intervention(s) were performed for stroke, or the neurological deficit results in death

AND

- No other readily identifiable non-stroke cause, e.g., progression of existing traumatic brain injury, seizure, tumor, metabolic or pharmacologic etiologies, is identified

AND

- Diagnosis is confirmed by neurology or neurosurgical specialist or neuroimaging procedure (MR, CT, angiography) or lumbar puncture (CSF demonstrating intracranial hemorrhage that was not present on admission)

ELEMENT VALUES

1. Yes
2. No

ADDITIONAL INFORMATION

- Onset of symptoms began after arrival to your ED/hospital.
- A diagnosis of stroke/CVA must be documented in the patient's medical record.
- Although the neurologic deficit must not present on admission, risk factors predisposing to stroke (e.g., blunt cerebrovascular injury, dysrhythmia) may be present on admission.

DATA SOURCE HIERARCHY GUIDE

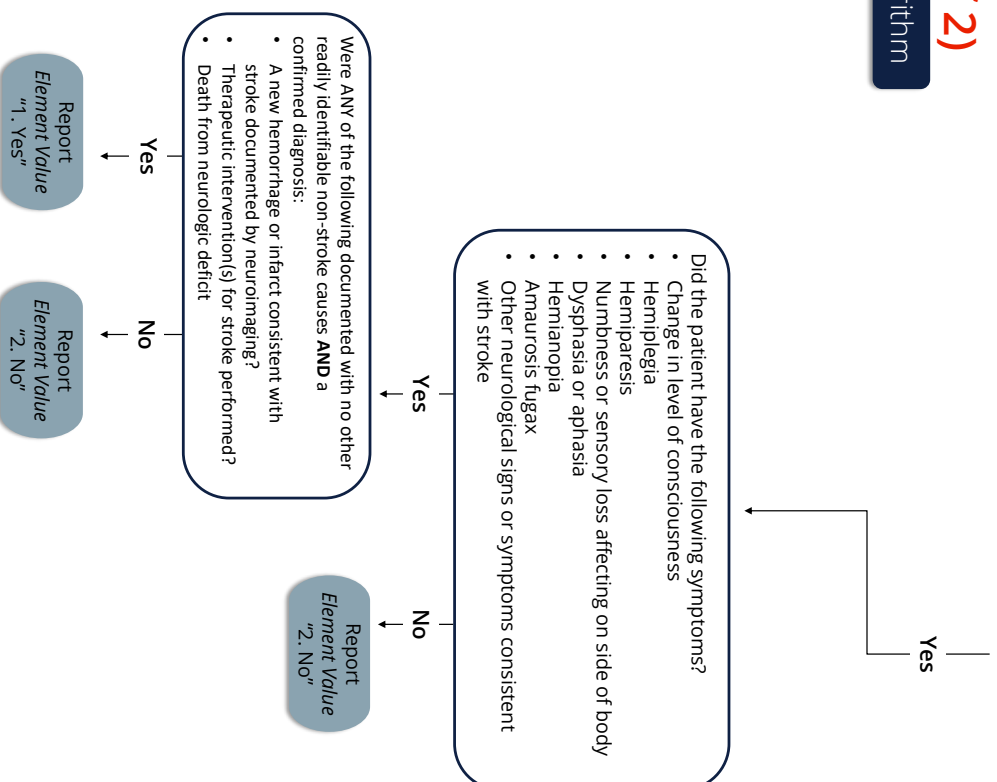
1. History and Physical
2. Physician Notes/Flow Sheet
3. Progress Notes
4. Case Management/Social Services Notes
5. Nursing Notes/Flow Sheet
6. Triage/Trauma Flow Sheet
7. Discharge Summary

ASSOCIATED EDIT CHECKS

Rule ID	Level	Message
20001	1	Value is not a valid menu option
20003	2	Element cannot be blank
20004	2	Element cannot be “Not Applicable”
20040	1	Single Entry Max exceeded

Stroke/CVA (pg. 2 of 2)

2025 NTDS Hospital Event Algorithm



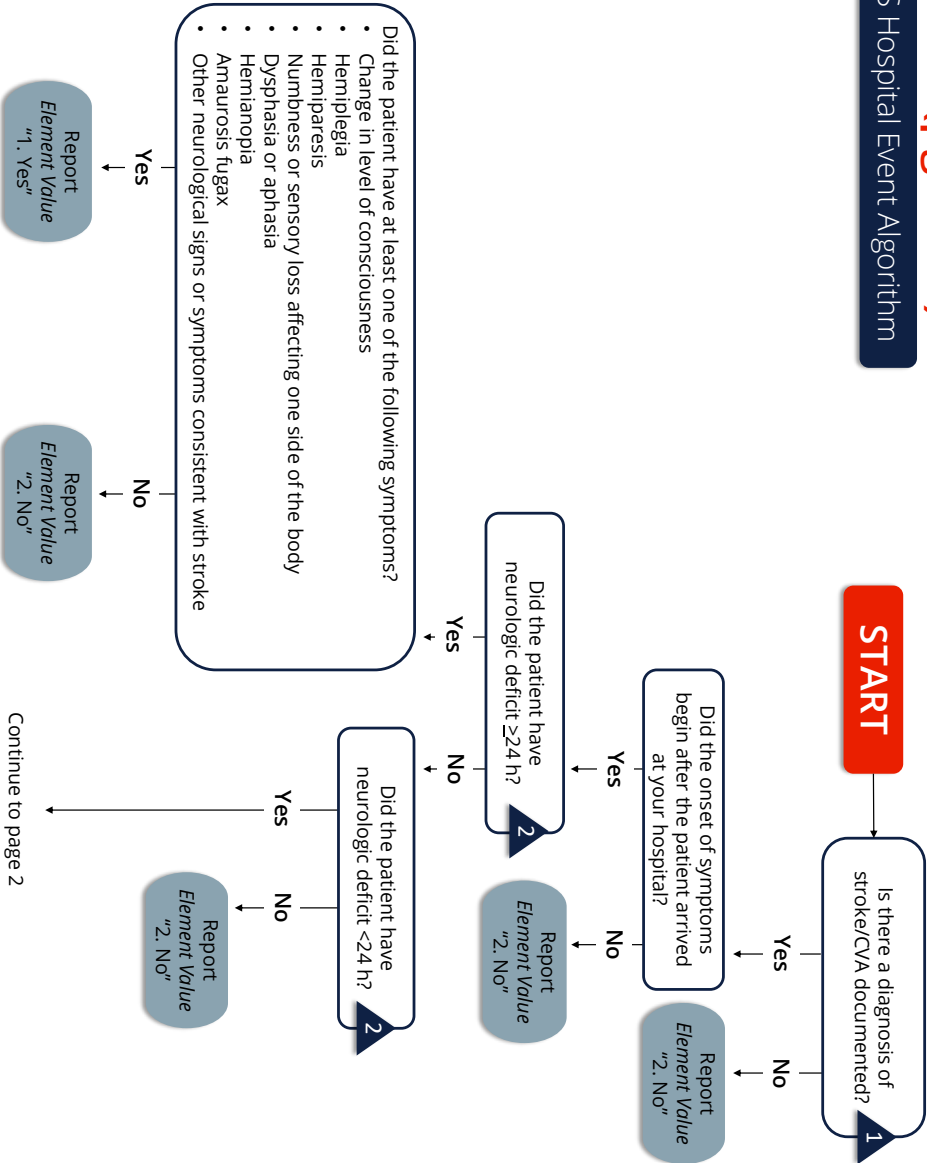
Consistent with the most recent version of the 2025 NTDS Data Dictionary
Refer to complete NTDS definition for additional information



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Stroke/CVA (pg. 1 of 2)

2025 NTDS Hospital Event Algorithm



1 A focal or global neurological deficit of rapid onset and NOT present on admission caused by a clot obstructing the flow of blood to the brain (ischemic stroke). Or by a blood vessel rupturing and preventing blood flow to the brain (hemorrhagic stroke). Or a transient ischemic attack which is temporarily caused by a temporary clot.

2 Although the neurologic deficit must not present on admission, risk factors predisposing to stroke (e.g., blunt cerebrovascular injury, dysrhythmia) may be present on admission.

Consistent with the most recent version of the 2025 NTDS Data Dictionary
Refer to complete NTDS definition for additional information

SUPERFICIAL INCISIONAL SURGICAL SITE INFECTION

DESCRIPTION

Must meet the following criteria:

Infection occurs within 30 days after any NHSN operative procedure (where day 1 = the procedure date)

AND

Involves only skin and subcutaneous tissue of the incision

AND

Patient has at least one of the following:

- a. Purulent drainage from the superficial incision.
- b. Organisms identified from an aseptically-obtained specimen from the superficial incision or sub cutaneous tissue by a culture or non-culture based microbiologic testing method which is performed for purposes of clinical diagnosis or treatment (e.g., not Active Surveillance Culture/Testing (ASC/AST)).
- c. Superficial incision that is deliberately opened by a surgeon, attending physician** or other designee and culture or non-culture based testing is not performed.

AND

Patient has at least one of the following signs or symptoms: pain or tenderness; localized swelling; erythema; or heat. A culture or non-culture based test that has a negative finding does not meet this criterion.

- d. Diagnosis of a superficial incisional SSI by the surgeon or attending physician** or other designee.

*The term attending physician for the purposes of application of the NHSN SSI criteria may be interpreted to mean the surgeon(s), infectious disease, other physician on the case, emergency physician, or physician's designee (nurse practitioner or physician's assistant).

COMMENTS: There are two specific types of superficial incisional SSIs:

1. Superficial Incisional Primary (SIP) – a superficial incisional SSI that is identified in the primary incision in a patient that has had an operation with one or more incisions (e.g., C- section incision or chest incision for CBGB).
2. Superficial Incisional Secondary (SIS) – a superficial incisional SSI that is identified in the secondary incision in a patient that has had an operation with more than one incision (e.g., donor site incision for CBGB).

ELEMENT VALUES

- | | |
|--------|-------|
| 1. Yes | 2. No |
|--------|-------|

ADDITIONAL INFORMATION

- Onset of symptoms began after arrival to your ED/hospital.
- A diagnosis of a surgical site infection must be documented in the patient's medical record.
- Consistent with the January 2019 CDC defined SSI.

DATA SOURCE HIERARCHY GUIDE

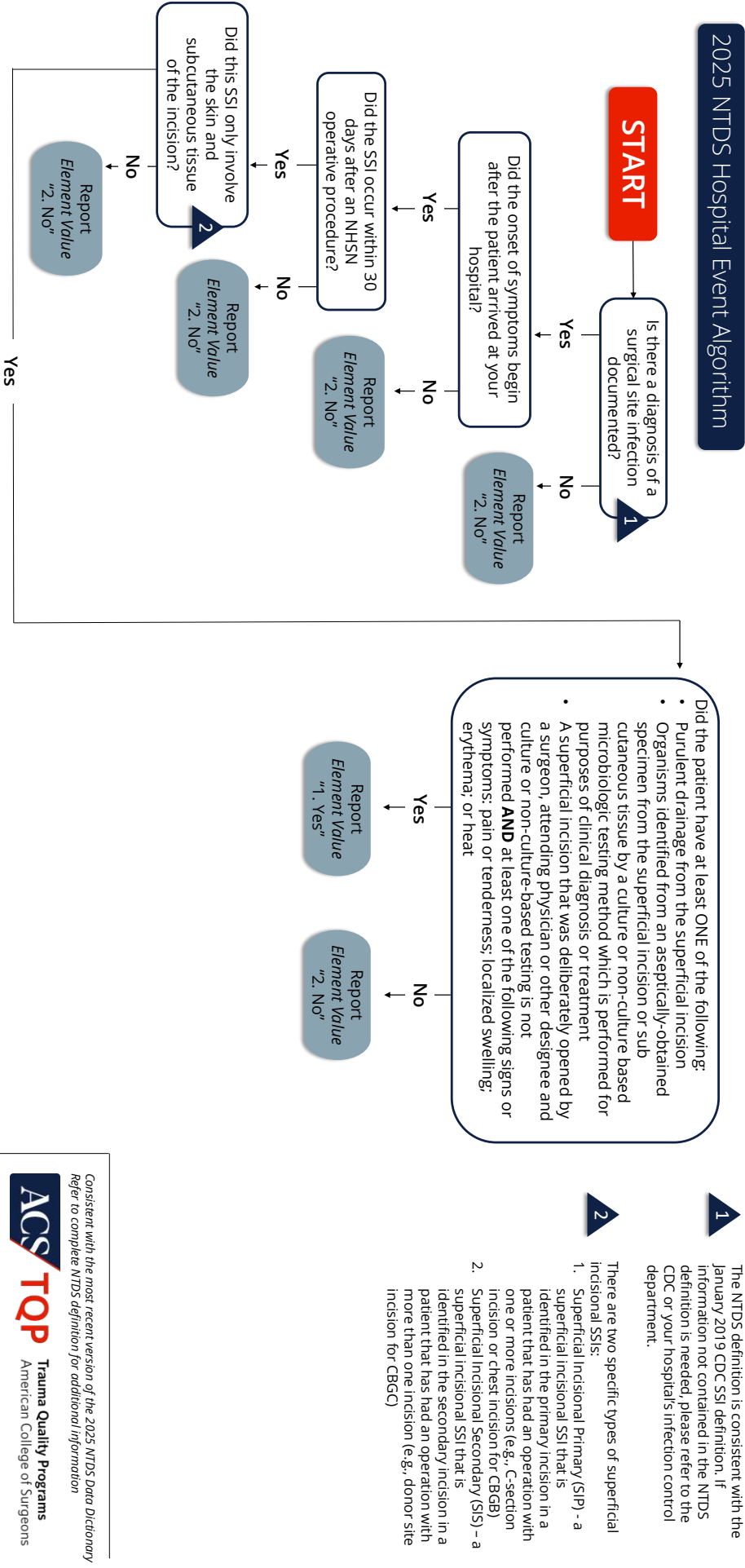
1. History and Physical
2. Physician Notes/Flow Sheet
3. Progress Notes
4. Case Management/Social Services Notes
5. Nursing Notes/Flow Sheet
6. Triage/Trauma Flow Sheet
7. Discharge Summary

ASSOCIATED EDIT CHECKS

Rule ID	Level	Message
20101	1	Value is not a valid menu option
20103	2	Element cannot be blank
20104	2	Element cannot be “Not Applicable”
20140	1	Single Entry Max exceeded

Superficial Incisional Surgical Site Infection (S/I SSI)

2025 NTDS Hospital Event Algorithm



Consistent with the most recent version of the 2025 NTDS Data Dictionary
Refer to complete NTDS definition for additional information

UNPLANNED ADMISSION TO THE ICU

DESCRIPTION

Patients admitted to the ICU after initial transfer to the floor, and/or patients with an unplanned return to the ICU after initial ICU discharge.

INCLUDE:

- Patients who required ICU care due to an event that occurred during surgery or in the PACU.

EXCLUDE:

- Patients with a planned post-operative ICU stay.

ELEMENT VALUES

1. Yes
2. No

ADDITIONAL INFORMATION

- Must have occurred during the patient's initial stay at your hospital.

DATA SOURCE HIERARCHY GUIDE

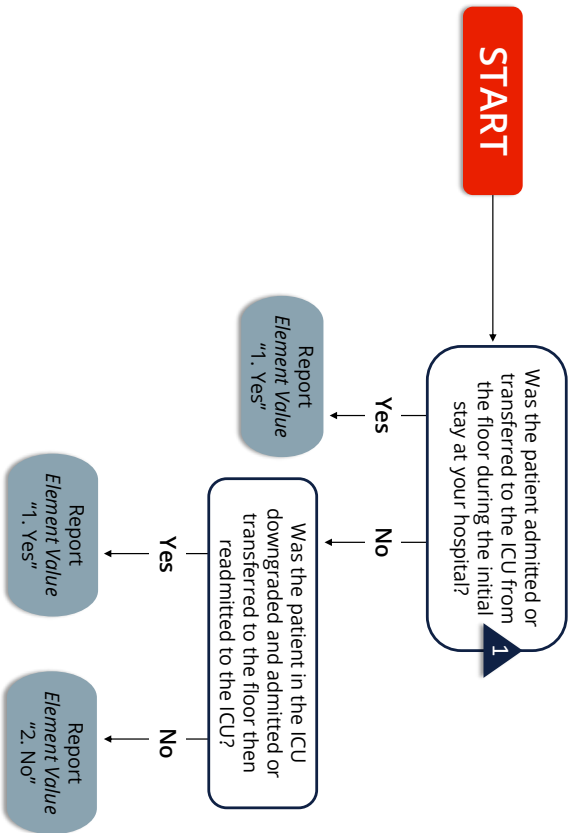
1. History and Physical
2. Physician Notes/Flow Sheet
3. Progress Notes
4. Case Management/Social Services Notes
5. Nursing Notes/Flow Sheet
6. Triage/Trauma Flow Sheet
7. Discharge Summary

ASSOCIATED EDIT CHECKS

Rule ID	Level	Message
20201	1	Value is not a valid menu option
20203	2	Element cannot be blank
20204	2	Element cannot be "Not Applicable"
20240	1	Single Entry Max exceeded

Unplanned Admission to the ICU

2025 NTDS Hospital Event Algorithm



1

Floor is defined as any other ward that is not an ICU. "Floor" also includes step-down units and the operating room for the purposes of this definition.

EXCLUDE: Patients with a planned post-operative ICU stay. This means that it was known prior to surgery that the patient would require post-operative ICU care.

INCLUDE: Patients who required ICU care due to an event that occurred during surgery or in the PACU.

Consistent with the most recent version of the 2025 NTDS Data Dictionary
Refer to complete NTDS definition for additional information



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UNPLANNED INTUBATION

DESCRIPTION

Patient requires placement of an endotracheal tube and mechanical or assisted ventilation manifested by severe respiratory distress, hypoxia, hypercarbia, or respiratory acidosis.

ELEMENT VALUES

1. Yes
2. No

ADDITIONAL INFORMATION

- Must have occurred during the patient's initial stay at your hospital.
- For patients who were intubated in the field or emergency department, or those intubated for surgery, an unplanned intubation occurs if they require reintubation > 24 hours after they were extubated.

DATA SOURCE HIERARCHY GUIDE

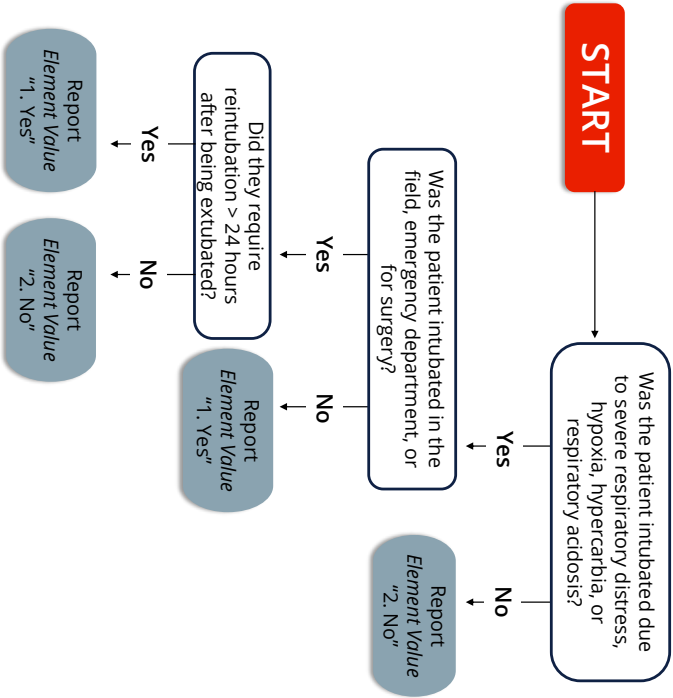
1. History and Physical
2. Physician Notes/Flow Sheet
3. Progress Notes
4. Case Management/Social Services Notes
5. Nursing Notes/Flow Sheet
6. Triage/Trauma Flow Sheet
7. Discharge Summary

ASSOCIATED EDIT CHECKS

Rule ID	Level	Message
20301	1	Value is not a valid menu option
20303	2	Element cannot be blank
20304	2	Element cannot be "Not Applicable"
20340	1	Single Entry Max exceeded

Unplanned Intubation

2025 NTDS Hospital Event Algorithm



Consistent with the most recent version of the 2025 NTDS Data Dictionary
Refer to complete NTDS definition for additional information

UNPLANNED VISIT TO THE OPERATING ROOM

DESCRIPTION

Patients with an unplanned operative procedure or patients returned to the operating room after initial operative management of a related previous procedure.

EXCLUDE:

- Non-urgent tracheostomy and percutaneous endoscopic gastrostomy.
- Pre-planned, staged and/or procedures for incidental findings.
- Operative management related to a procedure that was initially performed prior to arrival at your hospital.

ELEMENT VALUES

1. Yes
2. No

ADDITIONAL INFORMATION

- Must have occurred during the patient's initial stay at your hospital.

DATA SOURCE HIERARCHY GUIDE

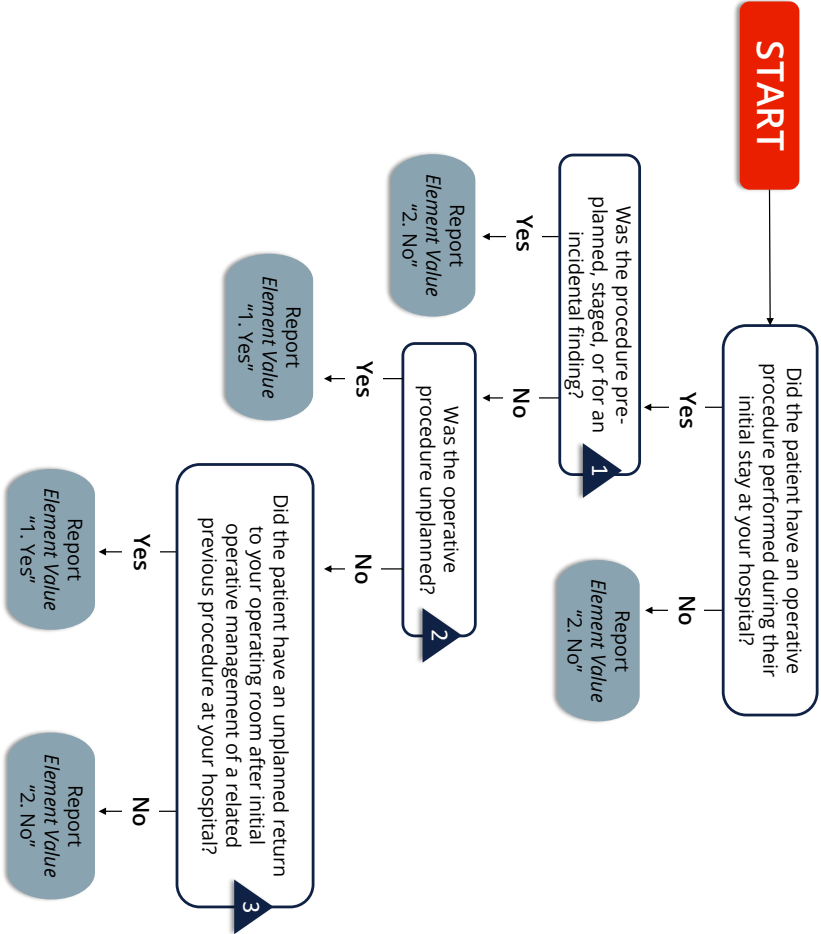
1. History and Physical
2. Physician Notes/Flow Sheet
3. Progress Notes
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5. Nursing Notes/Flow Sheet
6. Triage/Trauma Flow Sheet
7. Discharge Summary

ASSOCIATED EDIT CHECKS

Rule ID	Level	Message
21701	1	Value is not a valid menu option
21703	2	Element cannot be blank
21704	2	Element cannot be "Not Applicable"
21740	1	Single Entry Max exceeded

Unplanned Visit to the Operating Room

2025 NTDS Hospital Event Algorithm



1 A pre-planned procedure is a procedure indicated in the patient's original plan of care.

2 An unplanned procedure is a procedure that was not indicated in the patient's original plan of care.

EXCLUDE: Non-urgent tracheostomy and percutaneous endoscopic gastrostomy.

3 EXCLUDE: Operative management of related procedures performed prior to arrival at your hospital.

Consistent with the most recent version of the 2025 NTDS Data Dictionary
Refer to complete NTDS definition for additional information



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VENTILATOR-ASSOCIATED PNEUMONIA (VAP)

DESCRIPTION

A pneumonia where the patient is on mechanical ventilation for > 2 calendar days on the date of event, with day of ventilator placement being Day 1,

AND

The ventilator was in place on the date of event or the day before.

VAP Algorithm (*PNU2* Bacterial or Filamentous Fungal Pathogens):

IMAGING TEST EVIDENCE	SIGNS/SYMPTOMS	LABORATORY
Two or more serial chest imaging test results with at least <u>one</u> of the following:	At least <u>one</u> of the following:	At least <u>one</u> of the following:
• New and persistent or progressive and persistent • Infiltrate • Consolidation • Cavitation • Pneumatocoles, in infants ≤ 1 year-old NOTE: In patients <i>without</i> underlying pulmonary or cardiac disease (for example: respiratory distress syndrome, bronchopulmonary dysplasia, pulmonary edema, or chronic obstructive pulmonary disease), <u>one</u> definitive chest imaging test result is acceptable.	• Fever ($> 38^{\circ}\text{C}$ or $> 100.4^{\circ}\text{F}$) • Leukopenia ($< 4000 \text{ WBC/mm}^3$) or leukocytosis ($\geq 12,000 \text{ WBC/mm}^3$) • For adults ≥ 70 years old, altered mental status with no other recognized cause AND at least <u>one</u> of the following: • New onset of purulent sputum or change in character of sputum, or increased respiratory secretions, or increased suctioning requirements • New onset or worsening cough, or dyspnea, or tachypnea • Rales or bronchial breath sounds • Worsening gas exchange (for example: O_2 desaturations [for example: $\text{PaO}_2/\text{FiO}_2 < 240$], increased oxygen requirements, or increased ventilator demand)	• Organism identified from blood • Organism identified from pleural fluid • Positive quantitative culture or corresponding semi-quantitative culture result from minimally-contaminated LRT specimen (specifically, BAL, protected specimen brushing or endotracheal aspirate) • $\geq 5\%$ BAL-obtained cells contain intracellular bacteria on direct microscopic exam (for example: Gram's stain) • Positive quantitative culture or corresponding semi-quantitative culture result of lung tissue • Histopathologic exam shows at least <u>one</u> of the following evidences of pneumonia: – Abscess formation or foci of consolidation with intense PMN accumulation in bronchioles and alveoli – Evidence of lung parenchyma invasion by fungal hyphae or pseudohyphae

VAP Algorithm (PNU2 Viral, Legionella, and other Bacterial Pneumonias):

IMAGING TEST EVIDENCE	SIGNS/SYMPTOMS	LABORATORY
Two or more serial chest imaging test results with at least one of the following:	At least one of the following:	At least one of the following:
• New and persistent or progressive and persistent • Infiltrate • Consolidation • Cavitation • Pneumatocoles, in infants ≤ 1 year-old NOTE: In patients without underlying pulmonary or cardiac disease (for example: respiratory distress syndrome, bronchopulmonary dysplasia, pulmonary edema, or chronic obstructive pulmonary disease), <u>one definitive</u> chest imaging test result is acceptable.	• Fever ($> 38^{\circ}\text{C}$ or $> 100.4^{\circ}\text{F}$) • Leukopenia (< 4000 WBC/mm^3) or leukocytosis ($\geq 12,000$ WBC/mm^3) • For adults ≥ 70 years old, altered mental status with no other recognized cause AND at least one of the following: • New onset of purulent sputum or change in character of sputum, or increased respiratory secretions, or increased suctioning requirements • New onset or worsening cough, or dyspnea, or tachypnea • Rales or bronchial breath sounds • Worsening gas exchange (for example: O₂ desaturations [for example: PaO₂/FiO₂ < 240], increased oxygen requirements, or increased ventilator demand)	• Virus, <i>Bordetella</i>, <i>Legionella</i>, <i>Chlamydia</i> or <i>Mycoplasma</i> identified from respiratory secretions or tissue by a culture or non-culture based microbiologic testing method which is performed for purposes of clinical diagnosis or treatment (for example: not Active Surveillance Culture/Testing (ASC/AST)). • Fourfold rise in paired sera (IgG) for pathogen (e.g., influenza viruses, <i>Chlamydia</i>) • Fourfold rise in <i>Legionella</i> pneumophila serogroup 1 antibody titer to $\geq 1:128$ in paired acute and convalescent sera by indirect IFA. • Detection of <i>L. pneumophila</i> serogroup 1 antigens in urine by RIA or EIA

VAP Algorithm (PNU3 Immunocompromised Patients):

IMAGING TEST EVIDENCE	SIGNS/SYMPTOMS	LABORATORY
Two or more serial chest imaging test results with at least one of the following:	Patient who is immunocompromised (see definition in footnote) has at least one of the following:	At least one of the following:
• New and persistent or progressive and persistent • Infiltrate • Consolidation • Cavitation • Pneumatocoles, in infants ≤ 1 year-old NOTE: In patients without underlying pulmonary or cardiac disease (for example: respiratory distress syndrome, bronchopulmonary dysplasia, pulmonary edema, or chronic obstructive pulmonary disease), <u>one definitive</u> chest imaging test result is acceptable.	• Fever ($> 38^{\circ}\text{C}$ or $> 100.4^{\circ}\text{F}$) • For adults ≥ 70 years old, altered mental status with no other recognized cause • New onset of purulent sputum or change in character of sputum, or increased respiratory secretions, or increased suctioning requirements • New onset or worsening cough, or dyspnea, or tachypnea • Rales or bronchial breath sounds • Worsening gas exchange (for example: O_2 desaturations [for example: $\text{PaO}_2/\text{FiO}_2 < 240$], increased oxygen requirements, or increased ventilator demand) • Hemoptysis • Pleuritic chest pain	• Identification of matching <i>Candida</i> spp. from blood and one of the following: sputum, endotracheal aspirate, BAL or protected specimen brushing. • Evidence of fungi from minimally-contaminated LRT specimen (e.g., BAL or protected specimen brushing) from one of the following: – Direct microscopic exam – Positive culture of fungi – Non-culture diagnostic laboratory test OR • Any of the following from: LABORATORY CRITERIA DEFINED UNDER PNU2

VAP Algorithm ALTERNATE CRITERIA (PNU1), for infant's ≤ 1 year old:

IMAGING TEST EVIDENCE	SIGNS/SYMPTOMS/LABORATORY
Two or more serial chest imaging test results with at least one of the following:	Worsening gas exchange (for example: 2 desaturations [for example pulse oximetry < 94%], increased oxygen requirements, or increased ventilator demand)
• New and persistent or progressive and persistent • Infiltrate • Consolidation • Cavitation • Pneumatocoles, in infants ≤ 1 year-old NOTE: In patients without underlying pulmonary or cardiac disease (for example: respiratory distress syndrome, bronchopulmonary dysplasia, pulmonary edema, or chronic obstructive pulmonary disease), one definitive chest imaging test result is acceptable.	And at least three of the following: • Temperature instability • Leukopenia (≤ 4000 WBC/mm³) or leukocytosis (> 15,000 WBC/mm³) and left shift (> 10% band forms) • New onset of purulent sputum or change in character of sputum, or increased respiratory secretions or increased suctioning requirements • Apnea, tachypnea, nasal flaring with retraction of chest wall or nasal flaring with grunting • Wheezing, rales, or rhonchi • Cough • Bradycardia (< 100 beats/min) or tachycardia (> 170 beats/min)

VAP Algorithm ALTERNATE CRITERIA (PNU1), for children > 1 year old or ≤ 12 years old:

IMAGING TEST EVIDENCE	SIGNS/SYMPTOMS/LABORATORY
Two or more serial chest imaging test results with at least one of the following:	ALTERNATE CRITERIA, for child > 1 year old or ≤ 12 years old, at least three of the following:
• New and persistent or progressive and persistent • Infiltrate • Consolidation • Cavitation • Pneumatocoles, in infants ≤ 1 year-old NOTE: In patients without underlying pulmonary or cardiac disease (for example: respiratory distress syndrome, bronchopulmonary dysplasia, pulmonary edema, or chronic obstructive pulmonary disease), one definitive chest imaging test result is acceptable.	• Fever (> 38. 0°C or > 100. 4°F) or hypothermia (< 36. 0°C or < 96. 8°F) • Leukopenia (≤ 4000 WBC/mm³) or leukocytosis (≥ 15,000 WBC/mm³) • New onset of purulent sputum or change in character of sputum, or increased respiratory secretions, or increased suctioning requirements • New onset or worsening cough, or dyspnea, apnea, or tachypnea • Rales or bronchial breath sounds • Worsening gas exchange (for example: O₂ desaturations [for example pulse oximetry < 94%], increased oxygen requirements, or increased ventilator demand)

ELEMENT VALUES

1. Yes
2. No

ADDITIONAL INFORMATION

- Onset of symptoms began after arrival to your ED/hospital.
- A diagnosis of pneumonia must be documented in the patient's medical record.
- Consistent with the January 2019 CDC defined VAP.

DATA SOURCE HIERARCHY GUIDE

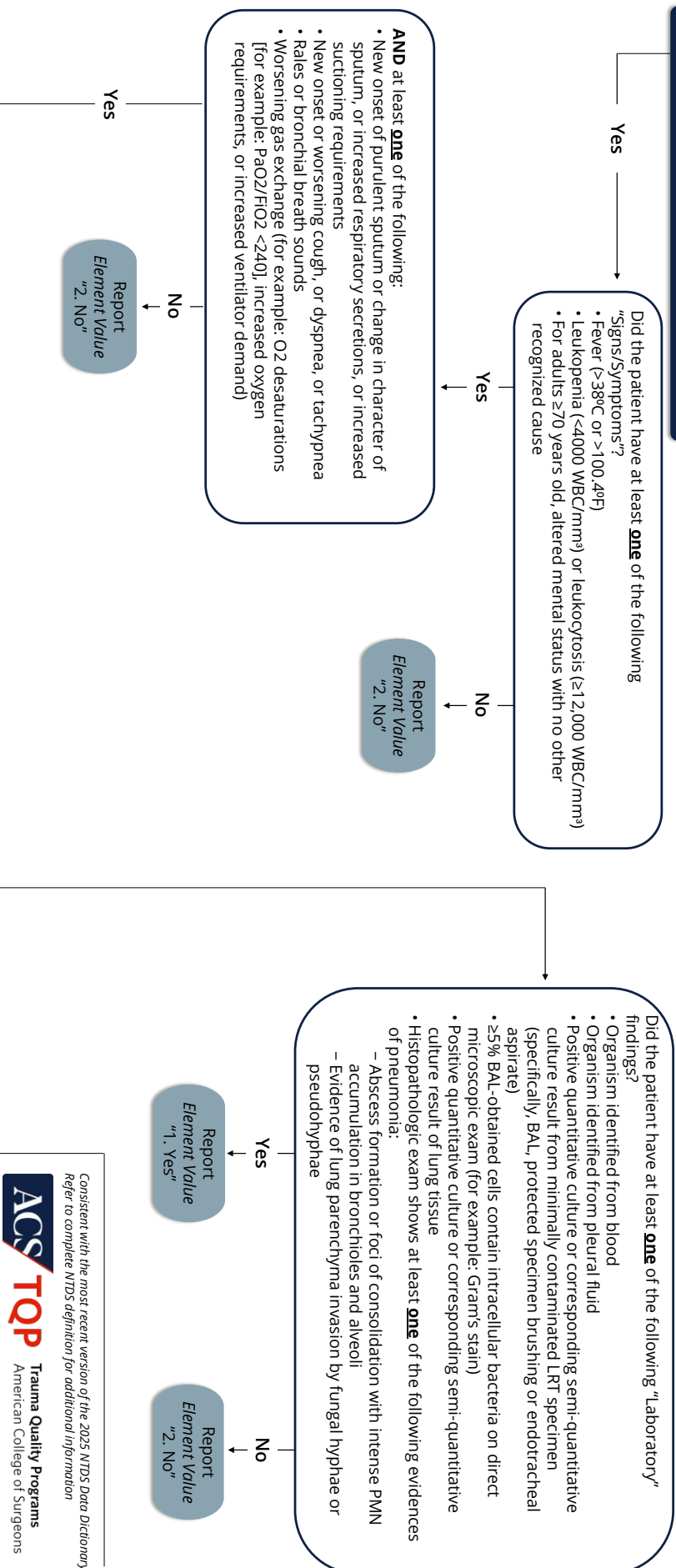
1. History and Physical
2. Physician Notes/Flow Sheet
3. Progress Notes
4. Case Management/Social Services Notes
5. Nursing Notes/Flow Sheet
6. Triage/Trauma Flow Sheet
7. Discharge Summary

ASSOCIATED EDIT CHECKS

Rule ID	Level	Message
20501	1	Value is not a valid menu option
20503	2	Element cannot be blank
20504	2	Element cannot be "Not Applicable"
20540	1	Single Entry Max exceeded

Ventilator-Associated Pneumonia (VAP) PNU2 Bacterial of Filamentous Fungal Pathogens (pg. 2 of 2)

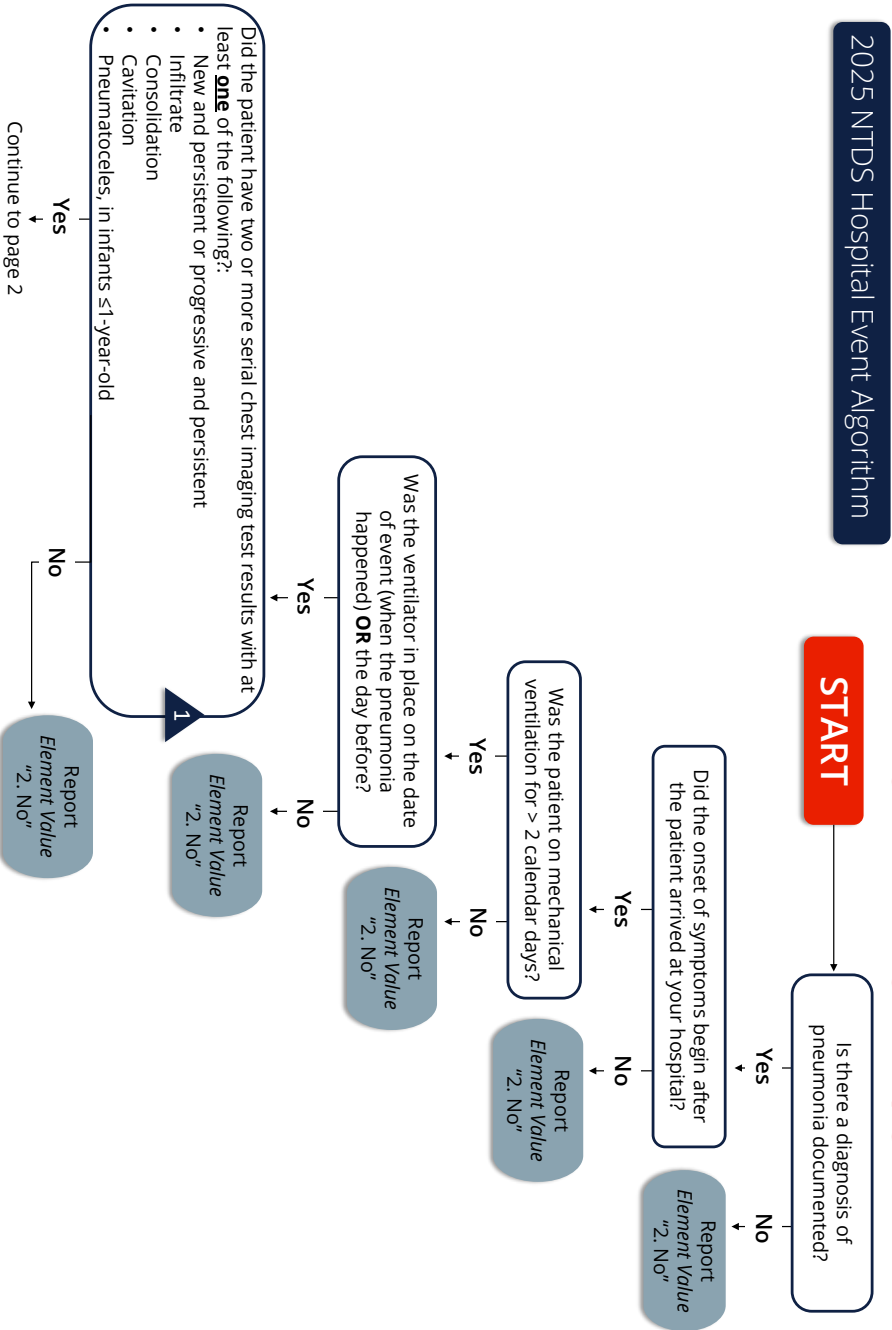
2025 NTDS Hospital Event Algorithm



Consistent with the most recent version of the 2025 NTDS Data Dictionary
 Refer to complete NTDS definition for additional information

Ventilator-Associated Pneumonia (VAP) PNU2 Bacterial of Filamentous Fungal Pathogens (pg. 1 of 2)

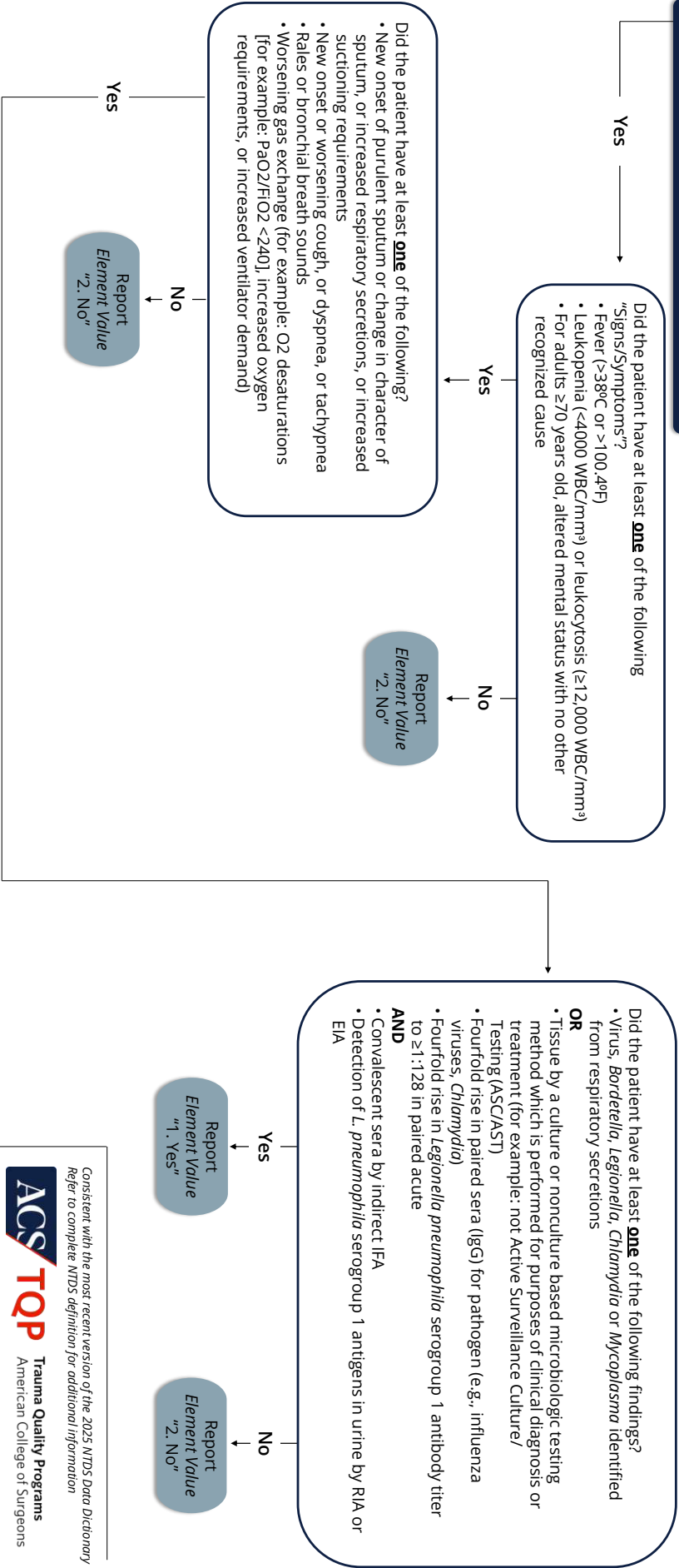
2025 NTDS Hospital Event Algorithm



1 NOTE: In patients without underlying pulmonary or cardiac disease (for example: respiratory distress syndrome, bronchopulmonary dysplasia, pulmonary edema, or chronic obstructive pulmonary disease), one definitive chest imaging test result is acceptable.

Ventilator-Associated Pneumonia (VAP) PNU2 Viral, Legionella, and other Bacterial Pneumonias (pg. 2 of 2)

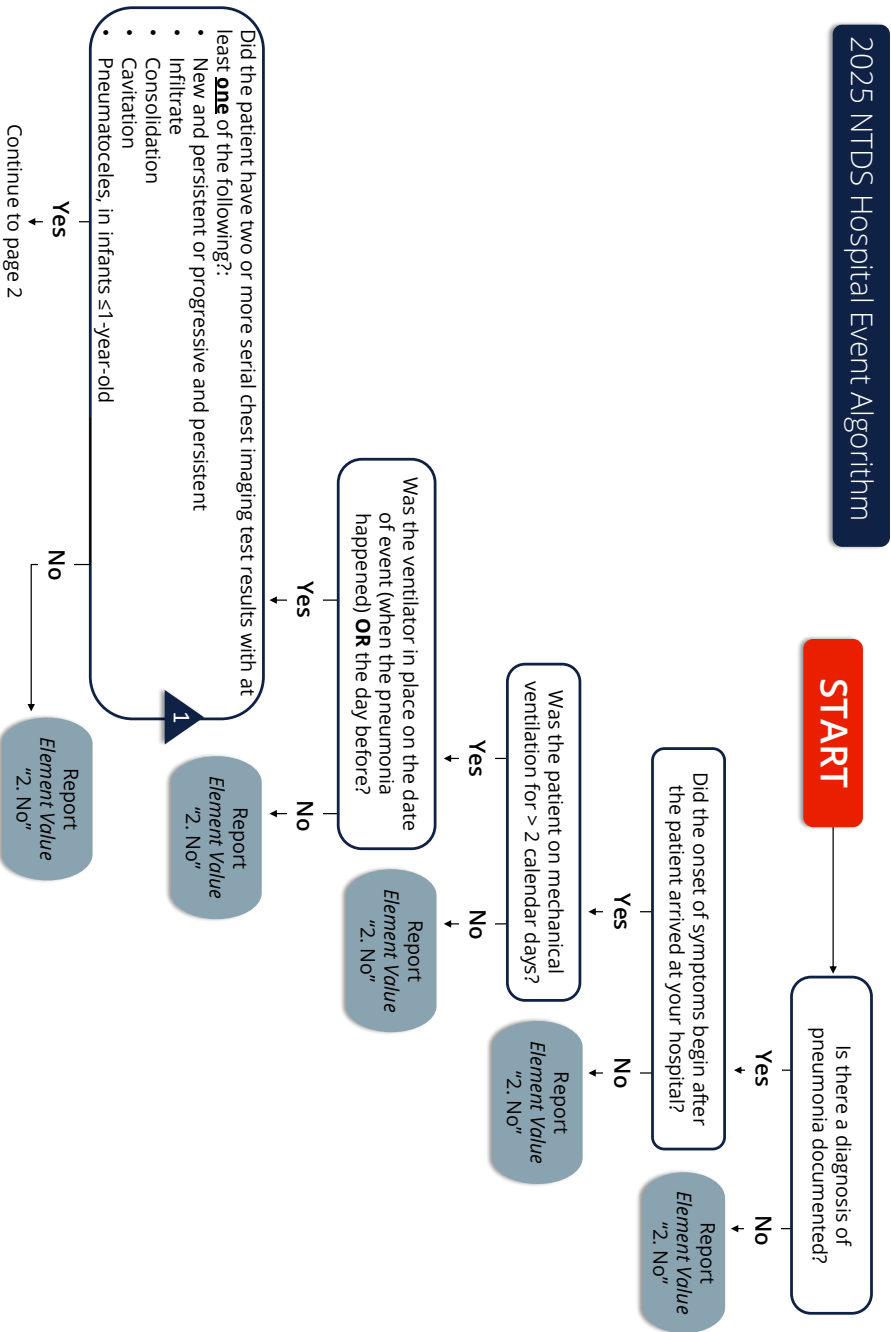
2025 NTDS Hospital Event Algorithm



Consistent with the most recent version of the 2025 NTDS Data Dictionary
Refer to complete NTDS definition for additional information

Ventilator-Associated Pneumonia (VAP) PNU2 Viral, Legionella, and other Bacterial Pneumonias (pg. 1 of 2)

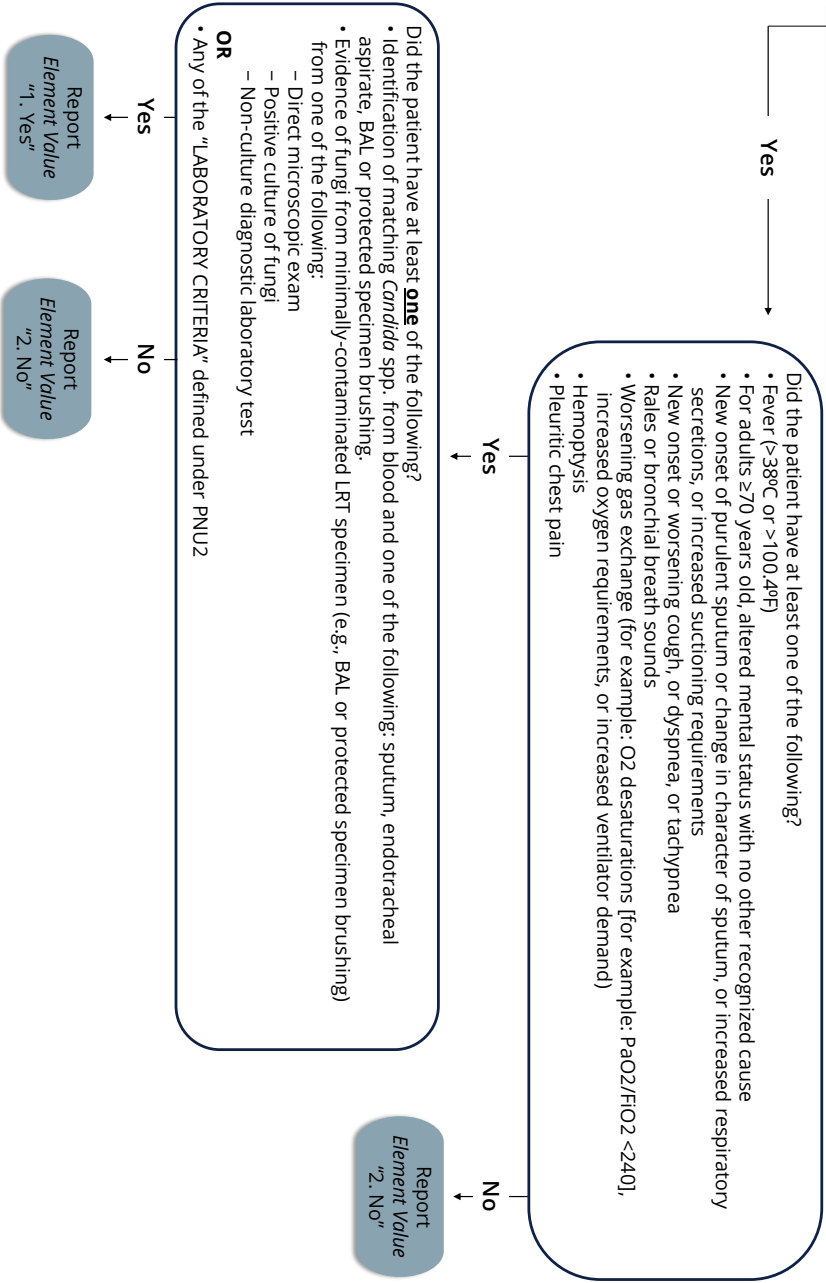
2025 NTDS Hospital Event Algorithm



NOTE: In patients without underlying pulmonary or cardiac disease (for example: respiratory distress syndrome, bronchopulmonary dysplasia, pulmonary edema, or chronic obstructive pulmonary disease), one definitive chest imaging test result is acceptable.

Ventilator-Associated Pneumonia (VAP) PNU3 Immunocompromised Patients (pg. 2 of 2)

2025 NTDS Hospital Event Algorithm

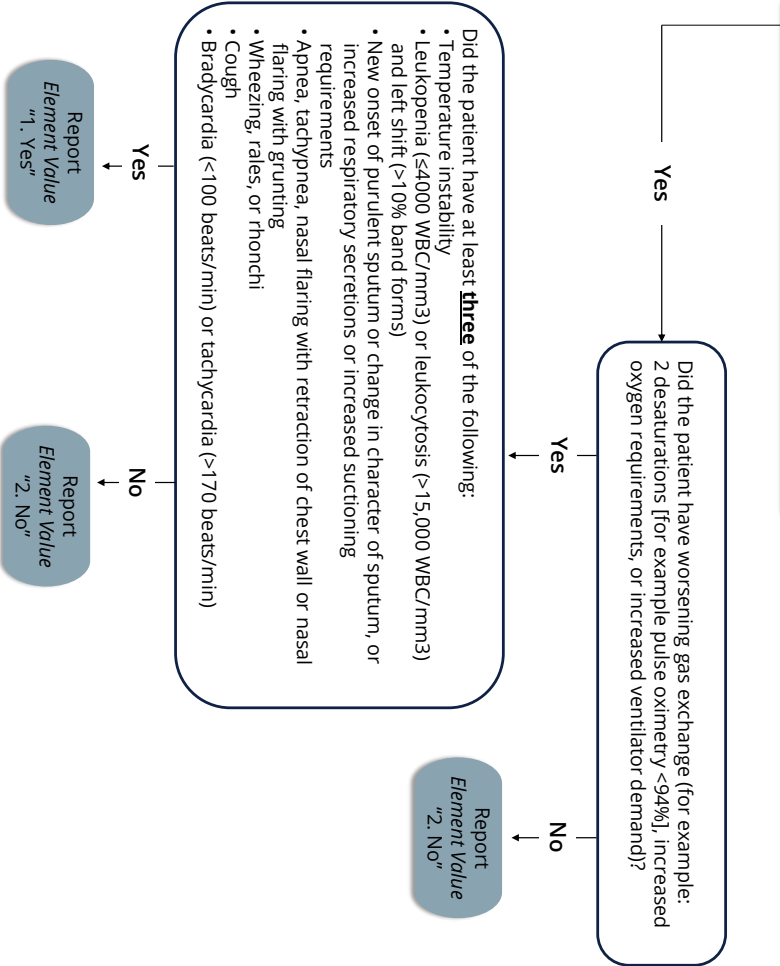


2025 NTDS Hospital Event Algorithm

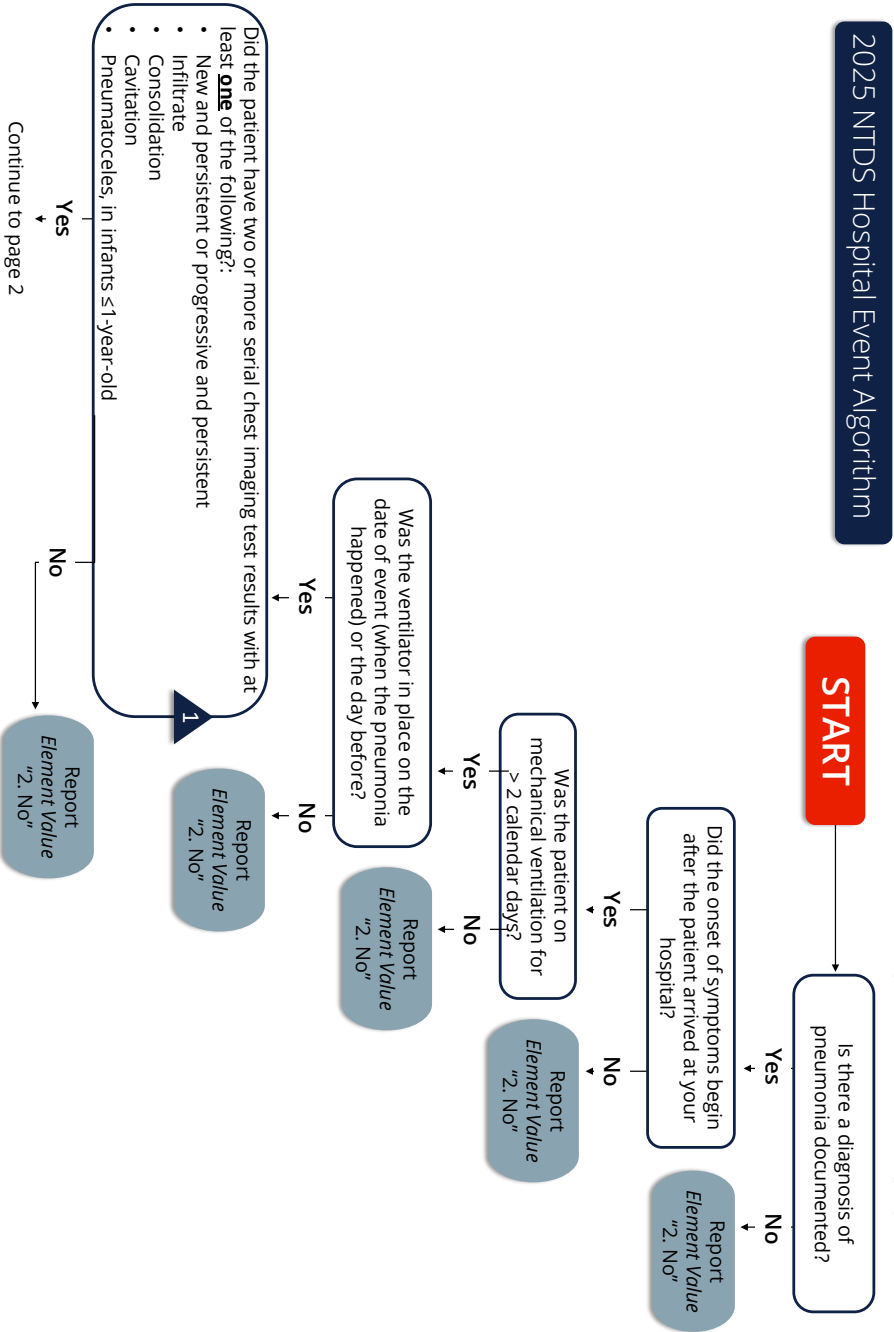


Ventilator-Associated Pneumonia (VAP) ALTERNATE CRITERIA (PNU1), for infants ≤ 1-year-old (pg. 2 of 2)

2025 NTDS Hospital Event Algorithm



Ventilator-Associated Pneumonia (VAP) ALTERNATE CRITERIA (PNU1), for infants ≤ 1-year-old (pg. 1 of 2)



NOTE: In patients without underlying pulmonary or cardiac disease (for example: respiratory distress syndrome, bronchopulmonary dysplasia, pulmonary edema, or chronic obstructive pulmonary disease), one definitive chest imaging test result is acceptable

2025 NTDS Hospital Event Algorithm

